

Pharmacology

World of Revision

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GENERAL PHARMACOLOGY : PART 1

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Types of Drugs

00:00:07

Orphan drugs : used for rare diseases → ↓ Profitability.

Essential drugs : meets healthcare needs of the majority of a population.

- Inexpensive.
- Easily available.
- Efficacious.
- Safe.
- Single molecule (Not fixed dose combination).

Prescription/legend drugs : Require prescription (Under schedule H).

Spurious drugs : Do not produce expected effect as drug component is falsified.

Misbranded drugs : Incorrect or missing information on drug label (Produces adequate effect).

Adulterated drug : unwanted additive in drug (Cough syrup : Glycerine contaminated with diethylene glycol causes → Renal failure).

P-drug :

- 'Personal drug' for any disease.
- STEP criteria to choose P-drug :
 - Safe.
 - Tolerable.
 - Efficacy.
 - ↓ Price.

Rational Drug Use :

Use of right drug for right disease & patient; at right dose, duration & route with right dispensation & monitoring ("Right price" not included).

Pharmacokinetics and Pharmacodynamics

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Pharmacokinetics :

Movement of drug through the body (ADME) :

- Absorption.
- Distribution.
- Metabolism.
- Excretion.

Pharmacodynamics :

Drug induced changes in body via target (DRE : Drug receptor effect).

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Drug Absorption

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- m/c mechanism : **Passive diffusion**.
- $\uparrow$ Diffusion : **unionized drug** (Same pH of drug and medium) d/t $\uparrow$ **lipid solubility**.
- **maximum absorption** $\rightarrow$ **Small intestine** (Large surface area).

pK_a : pH where drug $\rightarrow$ **50% ionized** & **50% unionized**.

Oral Absorption :

Good oral absorption drugs :

Drugs with :

- **Small size**.
- $\uparrow$ **Lipid solubility**.

Poor oral absorption drugs :

- **Proteins** d/t large size.
- Drugs ending with :
 - **tide** (Octreotide).
 - **ase** (Asparaginase).
 - **mab** (Trastuzumab).

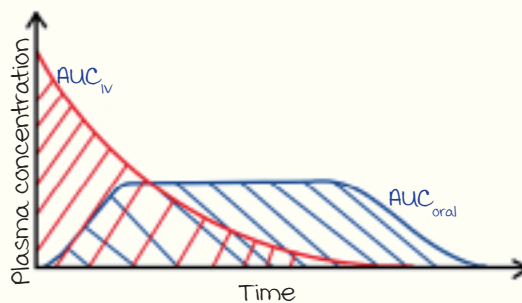
Extent and Rate of Absorption :

Extent of absorption :

- Amount of drug absorbed.
- AKA **bioavailability** (f : Fraction).
- Formula :

$$\text{Bioavailability (BA)} = \frac{AUC_{\text{oral}}}{AUC_{\text{iv}}}$$

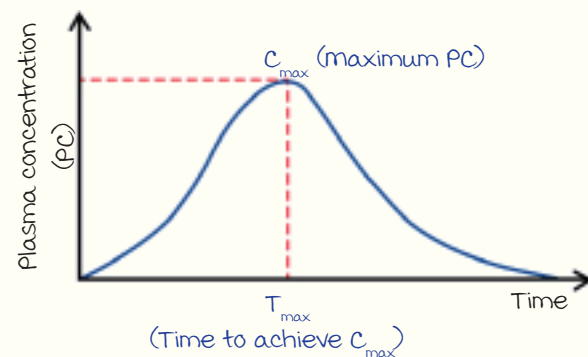
(AUC = Area under curve)



- Normal range of BA : **0 to 1**.
- 100% BA : IV and **inhalational gas**.
- BA depends on :
 - **Bypassing 1st pass metabolism**.
 - **Absorption**.

Rate of absorption :

- Amount of drug absorbed per unit of time.
- Determined by **T_{max}** .



Fastest rate : Inhalational route.

Role of ATP Binding Cassette (ABC) :

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- AKA p-glycoprotein (p-GP)/multidrug resistance-1 (MDR-1) pumps.
- Helps in drug efflux.

	Substance excreted by p-GP	Drugs & action on p-GP	Effect of drug
Intestinal cell	Digoxin	Clarithromycin : $\ominus$	Digoxin toxicity
		Rifampicin : $\oplus$	Digoxin failure
BBB	Loperamide	Quinidine : $\ominus$	Loperamide induced respiratory depression
Hepatocyte	Bile acids	Cyclosporine : $\ominus$	Cholestasis
Tumor cell/ bacteria	Anticancer/antibiotics (Cause resistance)	verapamil : $\ominus$ (Competitive)	$\ominus$ Development of resistance

Drug Distribution

00:41:47

Volume of Distribution (V_d) :

High : $V_d \geq 5L$	Low : $V_d < 5L$
$\uparrow$ Extravascular concentration Systemic circulation Tissues/organs (m/c: Adipose tissue)	$\uparrow$ Intravascular concentration Systemic circulation (Plasma) Tissues

 aV_d :

$$aV_d = \frac{D}{C_o} = \frac{\text{Dose of drug via IV route}}{\text{Initial PC } (C_{\max})}$$

$$D = aV_d \times C_T \text{ (Target PC)}$$

Significance :

I. Loading dose (LD) :

i. In IV route :

$$LD = aV_d \times C_T$$

ii. Other route :

$$D \times f = aV_d \times C_T$$

$$LD = \frac{aV_d \times C_T}{f}$$

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2. Dialysis : Not effective against high V_d drugs.

Drugs with $\uparrow V_d$ (BAD DOC)	Antidote
Benzodiazepine	Flumazenil
β -blocker	Glucagon
Amphetamines	Ammonium chloride
Digoxin	Digibind
Opioids	Naloxone
Organophosphates	Atropine
Calcium channel blockers	Calcium gluconate

Plasma Protein Binding :

Proteins :

Albumin (m/c)	Alpha-1-acid glycoprotein
Binds to acidic drugs	Binds to basic drugs
- Aspirin - Anti-coagulant (Warfarin) - Anti-epileptics/ anti-psychotics/ anti-depressants - Antibiotics (Sulfonamides)	- Opioids - Tricyclic anti-depressants β-blockers - Anti-arrhythmics (Amiodarone/Lidocaine)

Significance :

Hypoalbuminemia d/t :

i) $\downarrow$ Synthesis :

- $\downarrow$ Drug binding

↓
 $\uparrow$ Free drug

↓
 $\uparrow$ Toxicity.

- Seen in cirrhosis.

ii) $\uparrow$ Excretion :

- $\uparrow$ Drug excretion
(Albumin bound)

↓
 $\uparrow$ Failure.

- Seen in :
 - Nephrotic syndrome.
 - Diabetes mellitus.
 - Chronic kidney disease.

Drug Metabolism

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Phases :

	Phase I	Phase II (AKA conjugation)
mechanisms	Breakdown of drug (D) + Addition of functional group (FG)	Conjugate (-ve charged) binds to FG ↓ Ionised/water soluble drugs
Reactions	ORCHAD : - Oxidation (m/c) - Reduction - Cyclization - Hydrolysis - Aliphatic and aromatic hydroxylation - Deamination	GAMS (mnemonic) : - Glucuronidation (m/c) - Glycination - Glutathionation - Acetylation - Methylation - Sulfation
Enzyme involved	CYP450 enzymes : m/c : CYP3A4	Glucuronyl transferase (GT) : Glucuronidation
Clinical significance	-	Crigler Najjar syndrome : ↓GT → ↑Toxicity of : i) Irinotecan ii) Atazanavir

Note :

CYP450 enzymes (m/c : CYP3A4) :

- CY** : Cytochrome → Heme protein.
- P** : Pigments that absorb light of 450 nm wavelength.
- 3 : Family.
- A : Sub-family.
- 4 : Gene isoform number.

Drug-Enzyme Interaction :

	Enzyme inducers	Enzyme inhibitors
effect	Cause drug failure	Cause drug toxicity
Examples	mnemonic : GRAB PC - Griseofulvin - Rifampicin - Alcohol (Chronic consumption) - Benzopyrene - Phenytoin, Phenobarbital, Primidone - Carbamazepine, Cigarettes	mnemonic : QUICK VEG, DISK - Quinidine - Isoniazid, Protease inhibitors - Cimetidine, Chloramphenicol, Ciprofloxacin - Ketoconazole, Itraconazole, Fluconazole - Valproate - Erythromycin - Grapefruit juice - DEC, Delavirdine, Disulfiram
Important drug interactions	Rifampicin : - OCP failure - C/i in HIV with TB : - Affects Dolutegravir - Rx : Double the dose of Dolutegravir or change Rifampicin to Rifabutin	- Erythromycin → Theophylline toxicity - Clarithromycin → Statin toxicity

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Drugs metabolised by Plasma Esterase :Quick action of plasma esterase → Short $T_{1/2}$ of drugs.

Examples : Plasma Esterase Can Readily metabolise Short Acting drugs.

- Procaine, cocaine.
- Esmolol, Landiolol
- Clevidipine.
- Remifentanyl, Remimazolam.
- mivacurium.
- Succinylcholine.
- Acetylcholine.

Drug Excretion

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- m/c organ : Kidney.
- Differing pH b/w drug & medium → ↑ Ionization → ↑ Water solubility → ↑ Excretion.

Significance :

Drug toxicity :

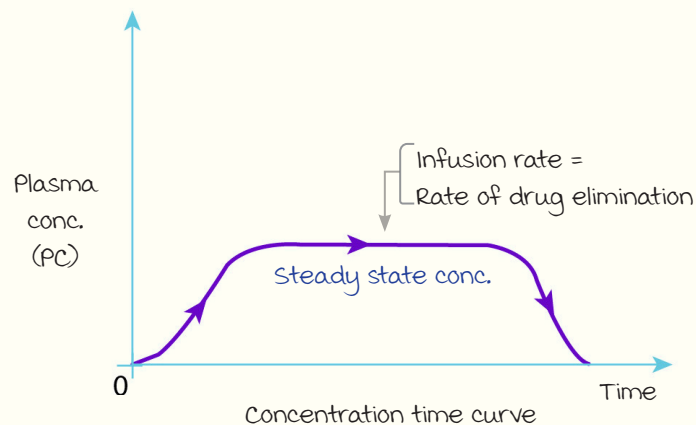
- Acidic drugs (Aspirin, Phenobarbital) → Alkalinisation of urine with bicarbonate.
- Basic drugs (Amphetamines) → Acidification of urine with ammonium chloride.

mechanisms :

Tubular secretion (80%) : Free + plasma protein bound. > Filtration (20%) : Free drug only.

Calculations :

Aim : Achieve & maintain steady state plasma concentration (SSPC).



Rate of drug elimination : Amount of drug excreted per unit of time.

$$\text{Rate (in mg/hr)} = PC \times \text{clearance}$$

Infusion rate (IR) : Amount of drug to be administered per unit of time
($\propto$ Clearance).

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$$IR = \text{Rate of drug elimination} = PC \times \text{clearance}$$

maintenance dose : Dose required to maintain steady state of drug PC
($\propto$ Clearance).

$$MD = PC \times \text{clearance} \times \text{time}$$

Half-life :

$$T_{1/2} = \frac{0.693 \times V_d}{\text{Clearance}} = \frac{0.693}{k_{\text{elimination}}}$$

$$(k_{\text{elimination}} = \frac{\text{Clearance}}{V_d})$$

Note :

Steady state plasma concentration (SSPC) :

- Achieved after $5 T_{1/2}$ s $\rightarrow$ Start therapeutic drug monitoring (TDM).
- E.g. : $T_{1/2}$ of lithium is 24 hours $\rightarrow$ TDM on 6th day.

Order of kinetics :

		Zero order kinetics	First order kinetics
Definition		Constant amount eliminated per hour	Constant proportion eliminated per hour
Effect of $\uparrow$ ing dose	$T_{1/2}$	$\uparrow$	constant
	Clearance	$\downarrow$	constant
	P.C.	Disproportionate $\uparrow$	Proportionate $\uparrow$
Risk of toxicity on overdosing		Higher	Lower
Examples		Zero ATP Has made Weak : • Alcohol • Theophylline, Tolbutamide • Phenytoin • Heparin • methanol • Warfarin	most other drugs

GENERAL PHARMACOLOGY : PART 2

Parameters of Pharmacodynamics

00:00:10

Drug $\xrightarrow{\text{Binds to}}$ Receptor $\xrightarrow{\text{Causes}}$ Effect.

Affinity :

- Tendency of a drug to bind to its receptor.
- marker of dose of drug $\left[\text{Affinity} \propto \frac{1}{\text{Dose}} \right]$.

Efficacy :

- maximum effect produced by a drug (most important factor).
- marker of effect of drug.

Potency :

- Relative dose of a drug required to produce particular effect.
- Potency $\propto \frac{1}{\text{Dose}}$.

Dose - Response Curve (DRC)

00:05:08

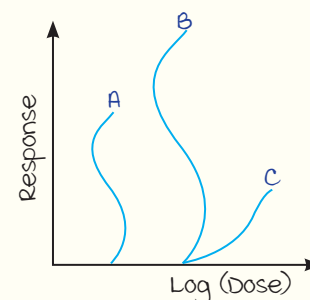
Graded DRC :

- Individual responses noted.
- Response is graded.

Efficacy : Height of curve
(B > A > C)

Potency $\propto 1/\text{Dose}$
(Left shift : A > B > C)

mnemonic : HELP.



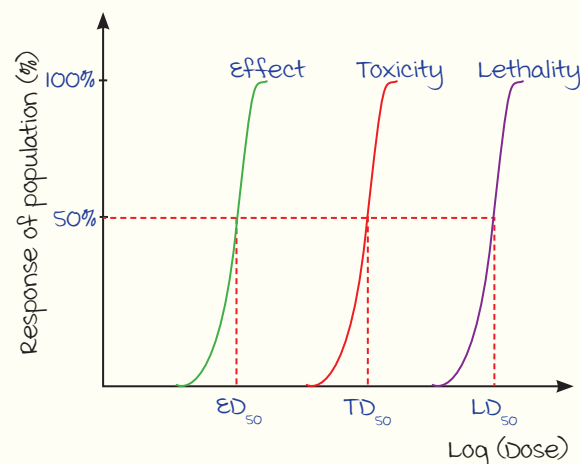
Affinity : Compared only if acting on same receptor.
(Parallel graph : A > B)

Quantal DRC :

- Population response noted.
- Response is binary (Yes/No) :
E.g. sedation.

ED₅₀ :

- Dose required to produce effect in 50% population.
- marker of potency.



TD_{50} : Dose required to produce toxicity in 50% population.
 LD_{50} : Dose required to kill 50% of animals.

markers of toxicity.

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Note :

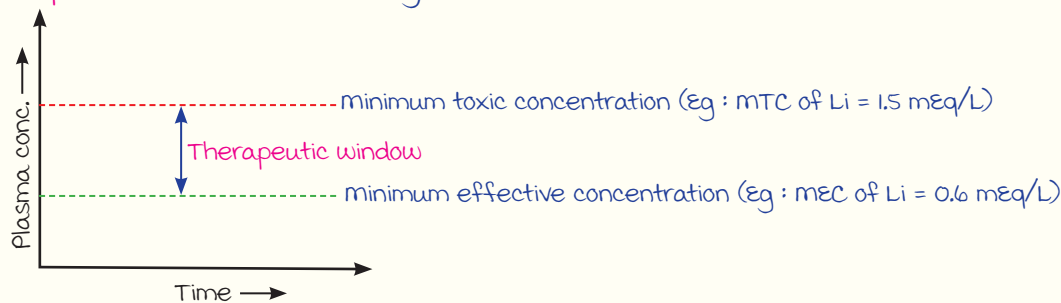
Lethality measured only in animals.

Therapeutic Index (TI) :

- marker of drug safety ($TI \propto$ Drug safety).
- 1. In humans : $TI = TD_{50} / ED_{50}$.
- 2. In animals : $TI = LD_{50} / ED_{50}$.
- Significance : Small change in dose $\rightarrow$ Toxicity (Eg : Lithium has $\downarrow$ TI).

Therapeutic Window/Range :

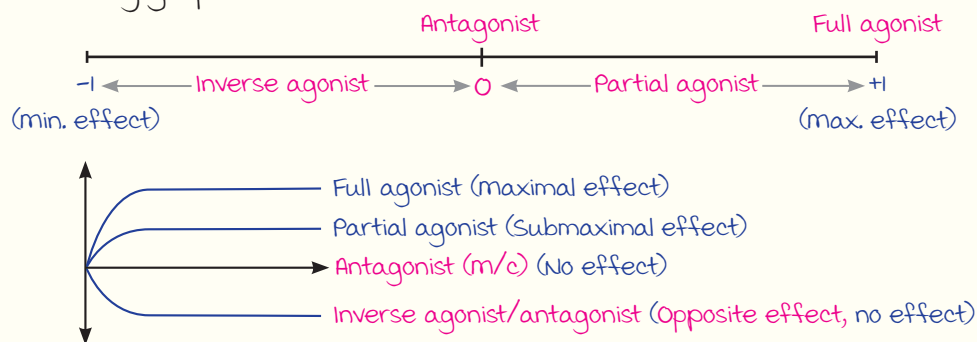
more important indicator of toxicity than TI.



Drug Receptor Interaction

00:20:46

Intrinsic efficacy graphs :



Type of Antagonism :

Physical/pharmacokinetic antagonism :

- Affected by physical binding of antagonist.
- E.g. : Charcoal in alcohol toxicity (Adsorbs alcohol).

Chemical antagonism :

E.g. :

- Heparin (Negative charge) : Protamine sulphate (Positive charge) $\rightarrow$ Neutralization
- Iron : Desferrioxamine (Chelation).

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Physiological antagonism :

- Antagonism at different receptors.
- E.g. :
 - Histamine via H_1 → Bronchoconstriction.
 - Adrenaline via β_2 → Bronchodilatation.

Competitive and non-competitive antagonism :

- Competitive antagonism
 - Irreversible :
 - Organophosphates.
 - Proton pump inhibitors.
 - Aspirin.
 - Reversible (m/c) : 99% of drugs.
- Competitive (Reversible) vs. non-competitive antagonism :

	Competitive (Reversible)	Non-competitive
Graph		
Change in graph	Right shift in DRC	↓ Height of DRC
Change in reaction	• V_{max} : Constant • Efficacy : Constant • K_m : ↑ • Potency : ↓	• V_{max} : ↓ • Efficacy : ↓ • K_m : Constant • Potency : Constant

Receptors

00:33:53

Types :

Type of receptor	Sub-types	Examples
Ligand gated ion channel	-	• GABA_A • Glutamate (NMDA, AMPA, Kainate) • Nicotinic • 5-HT₃
Enzymatic	Tyrosine Kinase receptors	• EGFR (Her-1) • Insulin, IGF-1 • Toll-like receptors • VEGFR • Her-2

Type of receptor	Sub-types	Examples
Enzymatic	Serine/threonine kinase receptors	TGFR
	Janus kinase receptors (JAK) : Sub-type of tyrosine kinase receptor	- Cytokine receptors (Eg : Leptin) - Prolactin - Growth hormone
	Guanylyl cyclase linked receptor	ANP & BNP (vasodilatation)
Nuclear	Located in nucleus (TRAP)	- Thyroid - Retinoic acid - Retinoid X - Estrogen - Progesterone - PPAR (Peroxisome Proliferator Activated Receptor)
	Located in cytoplasm	- mineralocorticoid - Glucocorticoid - Androgen - Vitamin D

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G-protein coupled receptors (GPCR) :

m/c subunit : α

GPCR	MOA	Examples	Drugs
G_s	$\oplus$ Adenylate cyclase $\uparrow$ cAMP Cardiac & skeletal muscles → Contraction Smooth muscles → Relaxation	β_1, β_a	-
G_q	$\oplus$ Phospholipase C $\uparrow$ IP₃ (2nd messenger) $\uparrow$ Ca²⁺ production Smooth muscle contraction.	α_p, m_p, m_3	- Oxytocin - Angiotensin
$G_{i/o}$ (Inhibitory)	$\downarrow$ cAMP (G_i) $\downarrow$ Ca²⁺ (G_o) - Open K⁺ channels (G_i and G_o) Relaxation	Presynaptic/ autoreceptors : m_a (Heart), $\alpha_2, H_3, 5-HT_1$	-
$G_{12/13}$	$\oplus$ Rho kinase Smooth muscle contraction	-	Rho Kinase $\ominus$: - Fasudil → Angina (Vasodilator) - Netarsudil → Glaucoma - Belumosudil → Immunosuppression

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Drug Safety

00:43:21

Pre-clinical trials : Only on animals.

Clinical Trials :

- On humans.
- Investigational New Drug (IND) : Drug under trial.

Phases of a clinical trial :

Phase	Features studied	Target population
I	• Toxicity • Safety • maximum tolerable dose • Pharmacokinetics, pharmacodynamics	• 20 - 100 healthy volunteers • 1 - 2 years • Open label trial
II	• Therapeutic exploratory trial/efficacy trial • Efficacy determined • Dose range/Dose safety • maximum drug failure	• 100 - 500 patients, unicentric • 2 - 3 years • Randomized control trial (RCT)
III	• Therapeutic confirmatory trial • Efficacy & safety confirmed ↓ Drug marketed.	• 500 - 3000 patients, multicentric • 3 - 5 years • Drug approved by : - CDSCO, India - FDA, USA • RCT/RUCT
IV	Post marketing surveillance (If ADR is rare/long term)	• many thousands of pt • No specific duration • Open label trial

Non-mandatory phases :

Phase 0 (microdosing) : 100 mcg of drug given, bound to radioligand (Assess pharmacokinetics).

Phase v (Pharmacoepidemiology) : Case control study, cohort study etc.

Note :

Phase I & phase II done together for toxic drugs like anti-cancer/anti-HIV.

- Patients are taken up directly.
- 20 to 100 patients involved.

Adverse Drug Reactions

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Types of ADR :

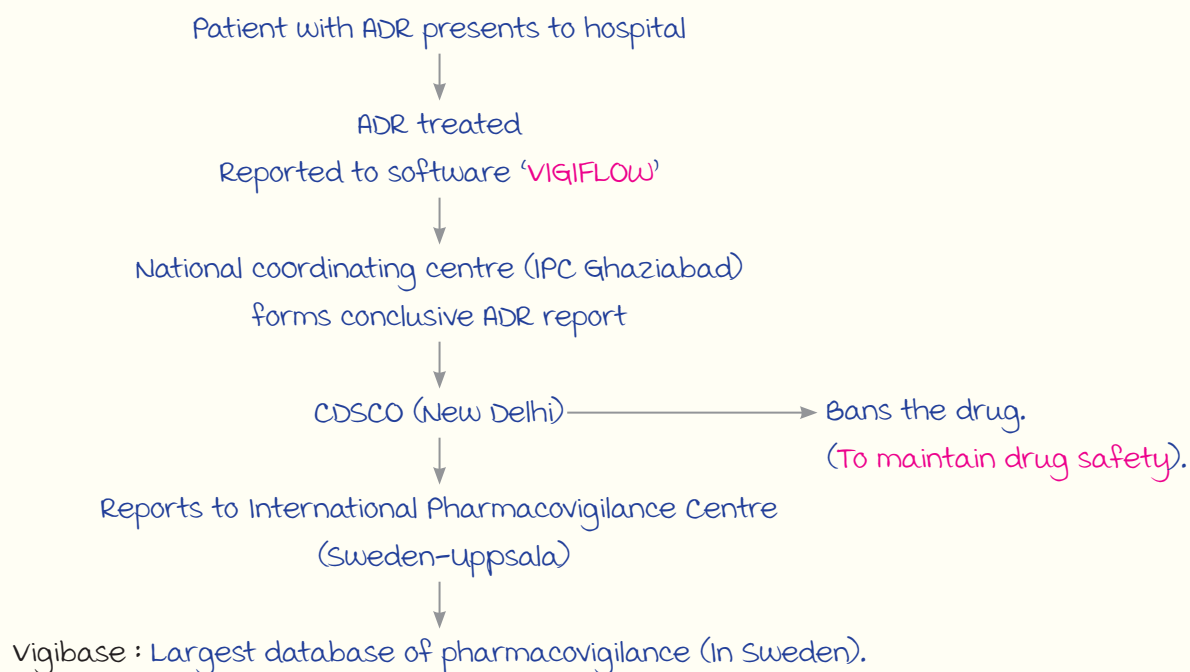
Mnemonic : **ABCDEF**.

Type	Features	Examples
A : Augmented	- Enhanced effect of drug - Dose dependent	Antihypertensive causing hypotension
B : Bizarre	Dose independent	Drug causing hypersensitivity (E.g. : Penicillin)
C : Chronic	Dose & duration dependent	Steroids → HPA suppression, Cushing syndrome
D : Delayed	Delay in time from exposure to ADR	Drugs causing teratogenicity (E.g. : NTDS d/t valproate)
E : End-of-use/ withdrawal	D/t stoppage of dose	- Opioid withdrawal - Clonidine withdrawal → HTN
F : Failure	Failure of drug	Enzyme inducers → OCP failure

Pharmacovigilance :

Sponsored by Govt. of India.

Flow of ADR information :



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Therapeutic Drug Monitoring

00:56:49

Principle : Plasma concentration (PC) correlates to effect/side effect.

Indications :

1. Clinical effect not easily quantified (If quantified → TDM not done :
 - Warfarin.
 - Antidiabetic drugs.
 - Anti-hypertensives).
2. Drugs having low therapeutic index :
 - Aminoglycosides.
 - Antiepileptics.
 - Lithium.
 - Theophylline.
 - Digoxin.
 - Cyclosporine.
 - Tacrolimus, tricyclic antidepressants.
3. Drugs with variable metabolism.
E.g. : metabolised by acetylation.
4. To check compliance :
E.g. : Antipsychotics.

Miscellaneous

00:59:17

Pharmacogenetics :

Mnemonic : A Malignant Gene Causes Very Severe Toxicity.

1. Acetyl transferase polymorphism :
 - NAT 1 gene : Fast acetylators (↓ Effect).
 - NAT 2 gene : Slow acetylators (↑ Toxicity).
2. Malignant hyperthermia :
D/t lignocaine, halothane, succinylcholine (m/c).
3. G-6-PD deficiency associated hemolysis : Dr. MANIS.
 - Dapsone.
 - methylene blue.
 - Antimalarial : Primaquine > Chloroquine > Quinine.
 - Nalidixic acid.
 - Nitrofurantoin.
 - Isoniazid.
 - Sulfonamides.

Drugs metabolized by acetylation

Mnemonic : HIPS Dance

- Hydralazine
- Isoniazid
- Procainamide
- Sulfonamides
- Dapsone

All can cause drug induced SLE

4. Glucuronyl transferase polymorphism : Causing irinotecan toxicity.

5. CYP450 enzyme polymorphism :

- CYP2C19 : Clopidogrel effect.
- CYP2C9 : variable effects of warfarin.
- CYP2D6 : ↑ Toxicity d/t psychiatric drugs.

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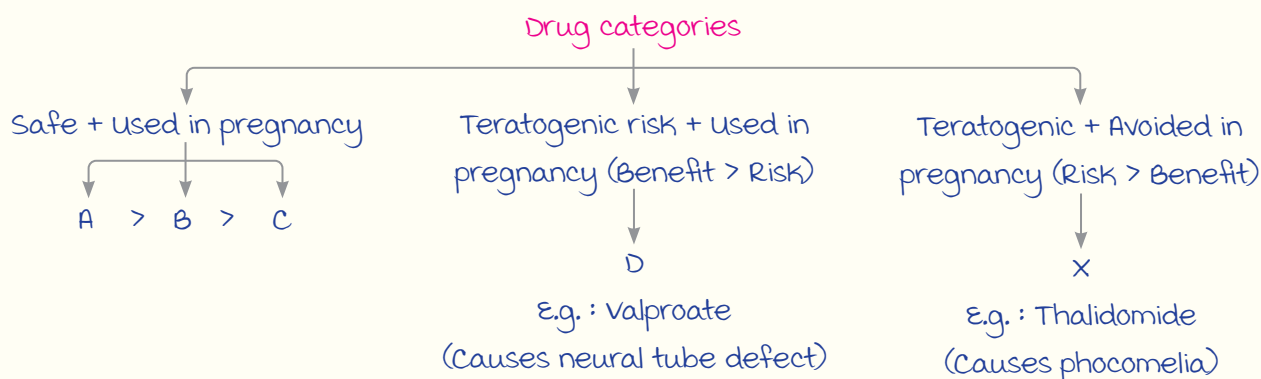
6. VKORC1 gene : variable metabolism of warfarin.

7. Succinylcholine associated prolonged apnea : D/t atypical pseudocholinesterase.

8. Thiopurine methyl transferase polymorphism :

Causing toxicity of Azathioprine, 6-mercaptopurine & 6-Thioguanine.

Pregnancy Drug Categories :



Drug Schedules :

	Features
Schedule G	Needs medical supervision
Schedule H	• Needs prescription • "Rx" in black color • "NRx" in red color (Risk of dependence)
Schedule H ₁ (2013)	"Rx" in red (Antibiotics)
Schedule X	• "XRx" in red color (Narcotic and psychotropic drugs : High potential for abuse). • Prescription copy : Retained by retailer for 2 years. • E.g. : - Amphetamine - Ketamine - methylphenidate

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DRUGS ACTING ON AUTONOMIC NERVOUS SYSTEM

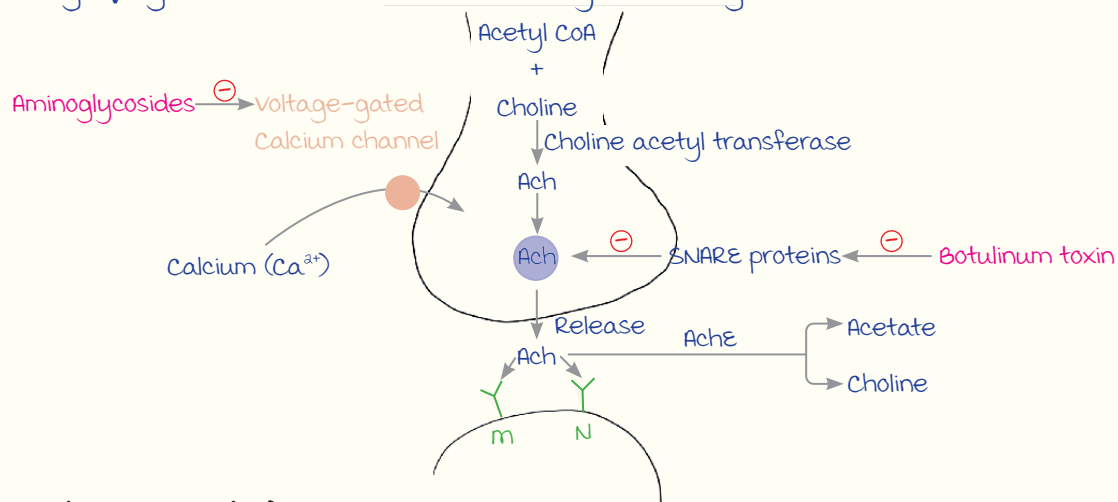
Regulation of ANS :

	Parasympathetic NS	Sympathetic NS
Receptors	muscarinic (m/c : m_3) or nicotinic	α or β
Neurotransmitter	Acetylcholine (ACh)	- Norepinephrine (NE) - Exceptions : - Dopamine $\xrightarrow{D_1}$ Renal blood vessels : vasodilation (Diuresis) - ACh in adrenals & sweat glands

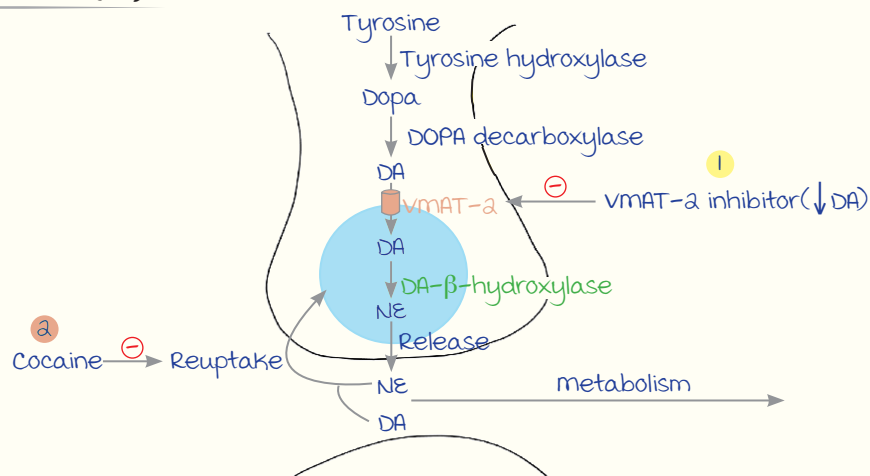
Neurotransmitters of ANS

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Acetylcholine (ACh) :

Drugs $\downarrow$ ing ACh $\rightarrow$ Neuromuscular toxicity (Breathing difficulties)

NE/Dopamine (DA) :



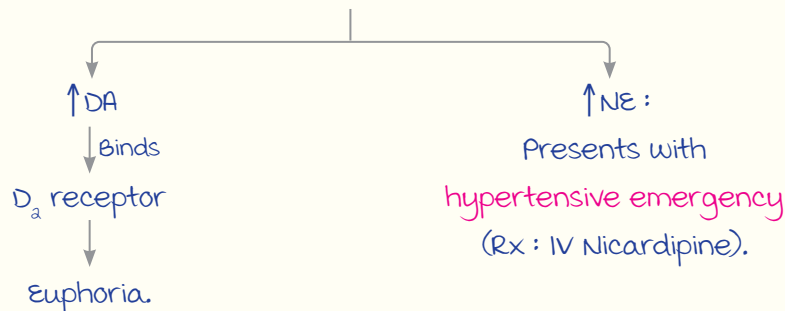
Drugs affecting NE/DA :

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• VMAT-2 inhibitors :

- Tetrabenazine  : DOC for Tics *a/w* Tourette syndrome.
Huntington's chorea.
- Deutetrabenazine, Valbenazine  : DOC for tardive dyskinesia.
(Longer acting)

• Cocaine :



- DOC : Cocaine dependence → Bromocriptine  (D₂ ⊕)

Systemic ANS Drugs Action

00:08:58

	Parasympathomimetic	Parasympatholytic	Sympathomimetic	Sympatholytic
CNS	• ↑ Cognition • Donepezil (DOC)  in Alzheimer's disease/dementia	• ↓ Cognition • Scopolamine : AKA truth serum for narcoanalysis (DOC : Thiopentone )	-	Lipid soluble β blockers s/e : • Depression • Insomnia • Nightmares
Pupil	Active miosis	Passive mydriasis	Active mydriasis	Passive miosis
	Pilocarpine (DOC)  : Closed angle glaucoma	Drugs : • Tropicamide • Atropine (1% ointment) • Homatropine • Cyclopentolate	Drugs : • Phenylephrine • Ephedrine	Drugs : • Phenoxybenzamine • Prazosin
		• Ocular fundus exam • In uveitis/corneal ulcers → Prevent synechiae • c/i : Closed angle glaucoma $\xrightarrow{\text{Ppt}}$ Acute congestive glaucoma		
		Cycloplegic function : • ↓ Iridocyclitis pain • Refractive error testing : - Children : 1% atropine - Adult : Tropicamide drops	No cycloplegic function	

----- Active space -----

	Parasympathomimetic	Parasympatholytic	Sympathomimetic	Sympatholytic
	↑	↓	-	-
Oro-pharyngeal secretions	Pilocarpine, Cevimeline (DOC) for xerostomia (Sjogren syndrome/ sicca syndrome)	Glycopyrrolate (DOC) (Does not cross BBB) > Atropine : As pre-anaesthetic medication (Prevents aspiration)	-	-
Bronchi	Bronchoconstriction C/i : Bronchial asthma (BA) & COPD	Bronchodilatation (M_3^-) For COPD : - Tiotropium (DOC) - Umeclidinium - Revefenacin LAMA (OD) Aclidinium → IAMA (BD) Ipratropium → SAMA (QID)	Bronchodilatation (β_2^+) β_2 agonists (S/e : Hyperglycemia) : - SABA : - Salbutamol - Terbutaline - Pirbuterol - LABA : - Formoterol - Salmeterol - vLABA : - Olodaterol - Vilanterol - Carmoterol SABA & formoterol are relievers (Fast acting)	Bronchoconstriction β blockers. - C/i : Bronchial asthma and COPD
Heart	↓ HR ↓ AV conduction	↑ HR ↑ AV conduction Atropine (DOC) : - AV block - Bradycardia (max. 3 mg) 1 mg $\xrightarrow{3-5 \text{ min}}$ 1 mg $\xrightarrow{3-5 \text{ min}}$ 1 mg • Low dose atropine < 0.5 mg (S/e : Bradycardia)	↑ HR ↑ AV conduction ↑ Contraction 1. Epinephrine : - Bradycardia (In children) - Cardiac arrest 2. Dobutamine : Acute CHF	↓ HR ↓ AV conduction ↓ Contraction β blockers (DOC) : - Atrial fibrillation/ Flutter (For rate control) - HOCm - Aortic dissection
Bladder detrusor	Contraction (M_3^+)	Relaxation (M_3^-)	- Relaxation (β_3^+) - Bladder sphincter : Contracts (α)	-

SABA : Short acting beta agonist
 LABA : Long acting beta agonist
 vLABA : Very long acting beta agonist

----- Active space -----

	Parasympathomimetic	Parasympatholytic	Sympathomimetic	Sympatholytic
Bladder detrusor	Bethanechol , neostigmine : - Post-op urinary retention - Bladder atony/overflow incontinence	Overactive bladder (urge incontinence) : - Fesoterodine - Darifenacin - Oxybutynin - Solifenacin - Tolterodine, - Trospium (⊗ BBB) - S/e : Dementia (via m₁); Least in : Darifenacin /solifenacin, trospium	- Overactive bladder (urge incontinence) : mirabegron (DOC) (β₃ agonist) - Stress incontinence : Duloxetine (SNRI : α₁⊕)	-
Skeletal muscles	Contraction	Relaxation	Contraction	Relaxation
	Edrophonium (DOC) : - Tensilon test : For Dx of myasthenia gravis (MG) - myasthenic vs. cholinergic crisis (Dx)	NDMR (NM receptor blocker) : muscle relaxants	β _a agonists : - Tremor : S/e - Clenbuterol : Performance enhancer (Banned)	β-blockers : ↓ Exercise tolerance (Fatigue : Avoid in athletes)
	Neostigmine : Dx & Rx of MG Cobra bite NDMR reversal			
	Pyridostigmine (DOC) : mx of MG Pre-medication atropine before edrophonium/neostigmine : Prevent muscarinic s/e			

Drug action on blood vessels :

	Sympathomimetic	Sympatholytic
MOA	vasoconstriction (α-agonist)	vasodilation (α-blocker)
Uses	- Spinal anaesthesia induced hypotension : (DOC) ↓ HR (DOC) : Ephedrine - Normal HR (DOC) : Phenylephrine	Pre-op HTN in pheochromocytoma (DOC) : Phenoxybenzamine (Oral α⁻) - F/b β-blocker (Prevents arrhythmia)
	Postural hypotension (DOC) : midodrine	Phentolamine (iv) : DOC Cheese reaction - Clonidine withdrawal HTN Intra-op HTN in pheochromocytoma
	Cardiogenic, septic, neurogenic shock (NGS) : - Norepinephrine IV (DOC) - NGS : Add atropine (D/t ↓HR)	Nicardipine in pheochromocytoma : - Oral : Pre-op HTN - IV : Intra-op HTN
	Anaphylactic shock : Epinephrine (DOC) Im route : 0.3 to 0.5 mg (1 : 1000 dilution)	HTN with BPH (α ₁ ⁻) : Terazosin, doxazosin, prazosin (DOC : Scorpion bite)
		BPH DOC : (α _{1a} ⁻) : Tamsulosin, silodosin.
		β-blockers : HTN in young age (↓Renin)

----- Active space ----- Drug action on GIT :

	Parasympathomimetic	Parasympatholytic
MOA	• ↑ HCl secretion • ↑ Contraction	• ↓ HCl secretion • ↓ Contraction
uses	Neostigmine : • Gastroparesis • Post-operative ileus	Pirenzepine, telenzepine : Peptic ulcer disease (Not used) Anti-spasmodics : • Glycopyrrolate • Dicyclomine (m/c)  • Scopolamine

Side Effects of Drugs

00:34:51

Drugs	Symptoms	Treatment
Cholinergic poisoning	• miosis : Pin-point pupil (m/c) • ↑ Salivation, sweating • Involuntary urination/defecation • Bradycardia	• OP poisoning : Atropine (DOC)  + Pralidoxime : most specific (AChE reactivator). • Atropine only : Carbamate poisoning • Normalised respiratory secretions : most specific.
Anticholinergic poisoning (Atropine, datura)	• mydriasis • Dry mouth • Urine retention/constipation • Tachycardia • Hyperthermia, dry skin	DOC  : Physostigmine
β-blockers	• Bronchospasm • Bradycardia • Insomnia, nightmares • Blocks hypoglycemia symptoms • Exercise intolerance	Glucagon (Source of cAMP for the heart)
α-blockers	• Postural hypotension • Ejaculation abnormality • Floppy iris	-

β -blockers

00:37:51

----- Active space -----

Cardioselective β -blockers :

Mr. BEAN Cardiologist.

- 2nd > 3rd generation.
- metoprolol.
- Betaxolol, Bisoprolol.
- Esmolol.
- Atenolol, Acebutolol.
- Nebivolol $\rightarrow$ most cardioselective, antioxidant action (+).
- Celiprolol.

3rd generation β -blockers
 $\downarrow$
 vasodilation

β -blockers with :

1. α -blockers (vasodilatation) : Carvedilol, Labetalol (m/c) $\rightarrow$.
2. NO release : Nebivolol.
3. Calcium channel blocker : Carvedilol.

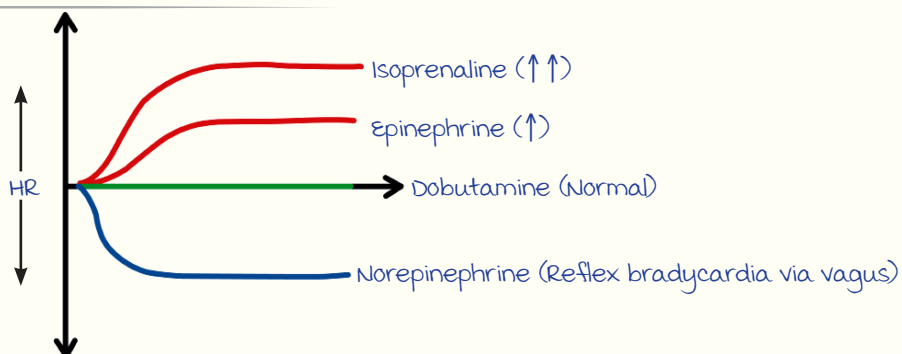
Important β -blockers :

Key features	β -blockers
Longest acting	Nadolol
Shortest acting	Landirolol > esmolol
Antioxidant	Nebivolol, Carvedilol
most cardioselective	Nebivolol
Partial agonist	Penbutolol, Pindolol, Acebutolol
Paralyses cell membrane (Sodium channel $\ominus$)	Propranolol, metoprolol

Physiological Effects of Autonomic Drugs

00:40:49

Effect of Catecholamines on HR :



Note :

NE with atropine/transplanted heart = $\uparrow$ HR.

----- Active space ----- Dopamine : use similar to NE but s/e : $\uparrow\uparrow$ HR

Dose dependent action  :

Continuous IV infusion.

Dose (mcg/kg/min)	Effect (Receptor)
> 0 - 2	Diuresis (β_1)
> 2 - 10	$\uparrow$ Contraction of heart (β_1)
> 10	Vasoconstriction (α_1)

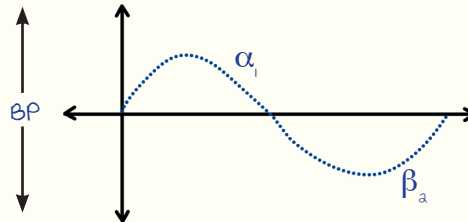
Epinephrine dilution :

Dilution	Route of administration
1 : 1000	- s/c - Im - Endotracheal
1 : 10,000	- IV - Intraosseous - Intracardiac (Not used)
1 : 100,000	- Local - Vasoconstriction
1 : 100,000 or 1 : 200,000	Local (with lignocaine)

Effect on Blood Pressure :

Dale's phenomenon :

Threshold concentration of epinephrine for $\alpha_1 > \beta_a$.

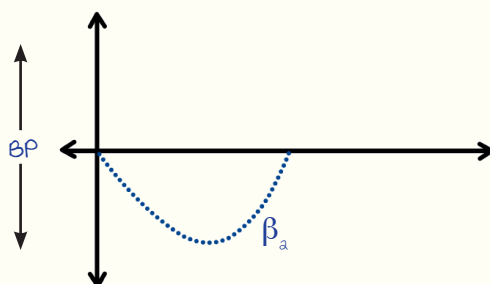


Biphasic response :

Initial high conc. : $\uparrow$ BP (α_1 mediated) $\xrightarrow[\text{excretion}]{\text{Drug metabolism}}$ Later low conc. : $\downarrow$ BP (β_a mediated).

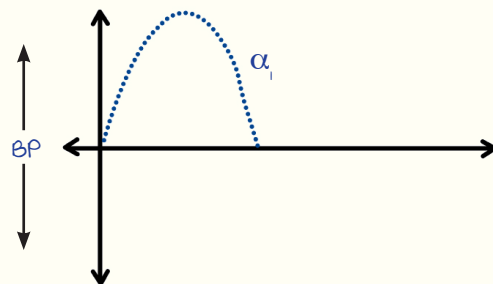
Vasomotor reversal of Dale :

- Epinephrine + α blocker in living system.
- Only $\downarrow$ in BP d/t unopposed β_a action.



Vasomotor re-reversal of Dale :

- Epinephrine + β -blocker in living system.
- Steep $\uparrow$ in BP d/t unopposed α_1 action.



Anti-glaucoma Drugs

00:50:21

----- Active space -----

	MOA	use	Side effects & contraindications
Prostaglandin analogs : • Latanoprost • Bimatoprost	↑ uveoscleral outflow	DOC  : Open angle glaucoma (OAG)	• Iris pigmentation, trichomegaly, dry/sandy eyes, cystoid macular edema • C/i : uveitis/herpetic keratitis
β-blockers : • Timolol, metipranolol, levobunolol, carteolol • Betaxolol : Preferred in asthma, COPD	↓ Aqueous production	and line : Open angle glaucoma	• Timolol : NLD obstruction • metipranolol : Granulomatous anterior uveitis • C/i : Systemic β-blockers (Total conduction block)
Epinephrine } Dipivefrin } Rare	↓ Aqueous production	Open angle glaucoma (↓ use)	• Ocular allergy • Black pigmentation of conjunctiva
α _a  : • Apraclonidine • Brimonidine	↓ Aqueous production	Open angle glaucoma	• Apraclonidine : Lid retraction, mydriasis, conjunctival blanching. • Brimonidine (Crosses BBB) : Apnea in neonates, drowsiness, hypotension
miotic agents : • Pilocarpine • Physostigmine • Echothiophate • Carbachol	↑ Trabecular outflow	Pilocarpine (DOC)  : In closed angle glaucoma	• Common s/e : - Brow ache - Accommodation spasm - Retinal detachment • Echothiophate : Iris cysts • Echothiophate/physostigmine : Cataract
Carbonic anhydrase  : • Oral/iv : Acetazolamide • Topical : Brinzolamide	↓ Aqueous production	Acetazolamide : Acute congestive glaucoma Brinzolamide : OAG	-
Rho Kinase  : Netarsudil	↑ Trabecular outflow	Open angle glaucoma	• Conjunctival hyperemia • Corneal verticillata

DRUGS ACTING ON CVS

Anti-Arrhythmic Drugs

00:00:20

Tachyarrhythmia :

- Atrial : SVT, PSVT, atrial fibrillation.
- Ventricular : VF, VT.
- Long QT syndrome, WPW syndrome.

Bradyarrhythmia :

DOC : Atropine.

SVT, PSVT :

- Arising from atrium.
- Aim : Block AV node (Prevent conduction into ventricles).

Drugs : ABCD

IV drugs :

- Acute attack
- ↓BP

• A : Adenosine

• B : Beta blocker

• C : CCB (Diltiazem/verapamil)

• D : Digoxin

Oral

(Chronic mx)

Order of preference

DOC :

- Acute : Adenosine.
- Long term mx : β blockers.

mx of acute attack :

↓BP (Unstable)

Cardioversion.

Normal BP (Stable)

Carotid massage (↑vagal)

H/o asthma/COPD

Absent

Diltiazem/verapamil.

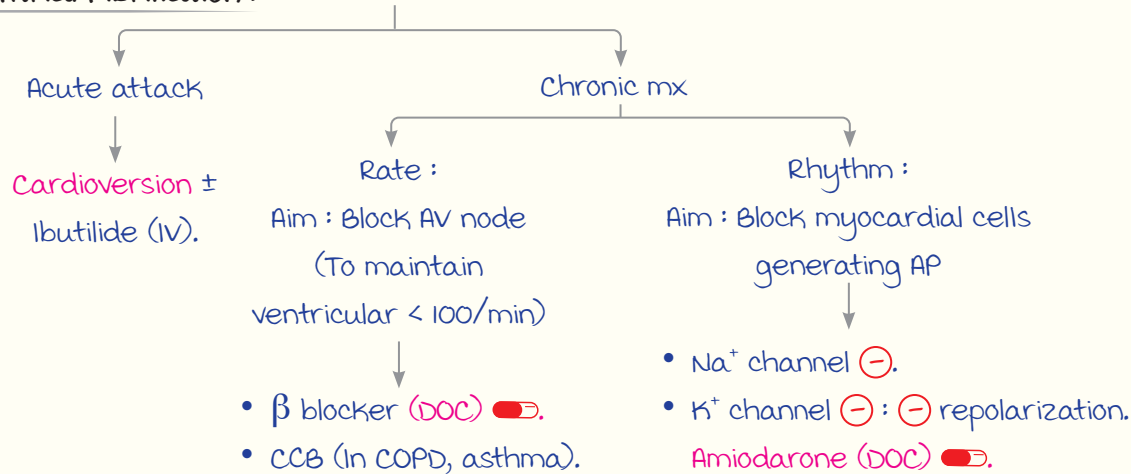
Adenosine (DOC) :

- IV rapid push.
- Adults : 6 mg.
- Children : 0.1 mg/kg.

Note :

- Adenosine, β blocker : S/e $\rightarrow$ Bronchoconstriction (C/i : Asthma).
- Adenosine : S/e $\rightarrow$ A. fibrillation.
- Digoxin indication : SVT/PSVT + Chronic CHF.

----- Active space -----

Atrial Fibrillation :WPW Syndrome :

Anatomical accessory pathway.

Rx :

- DOC : Flecainide (S/e : most arrhythmogenic) .
- TOC : Radiofrequency ablation.
- A/w atrial fibrillation : IV procainamide.

----- Active space -----

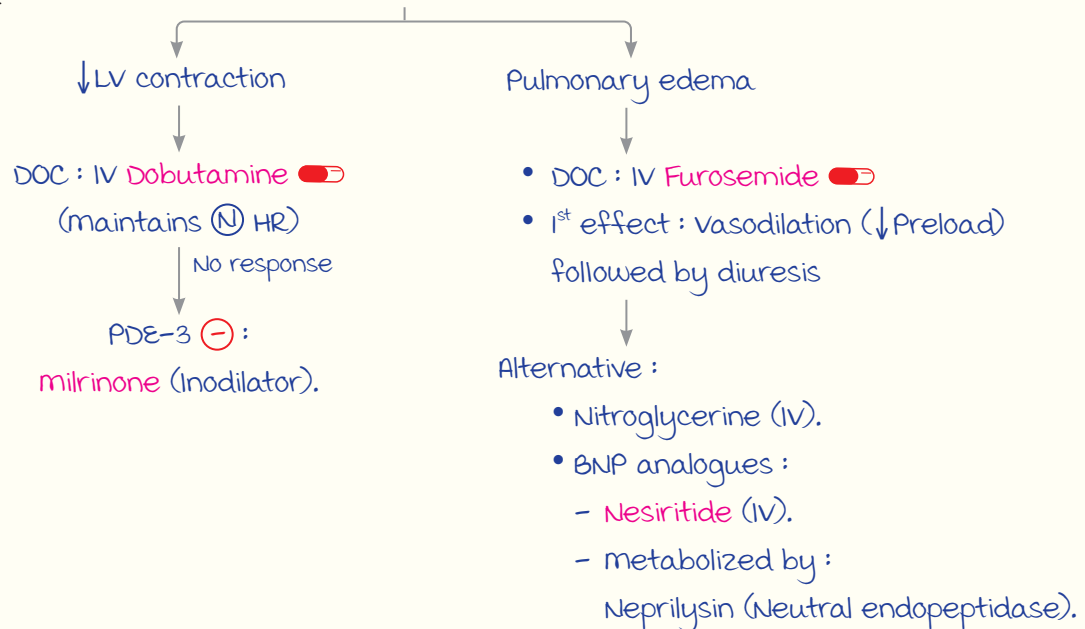
Heart Failure Drugs

00:20:20

Acute CHF :

m/c cause : MI (m/c of LV).

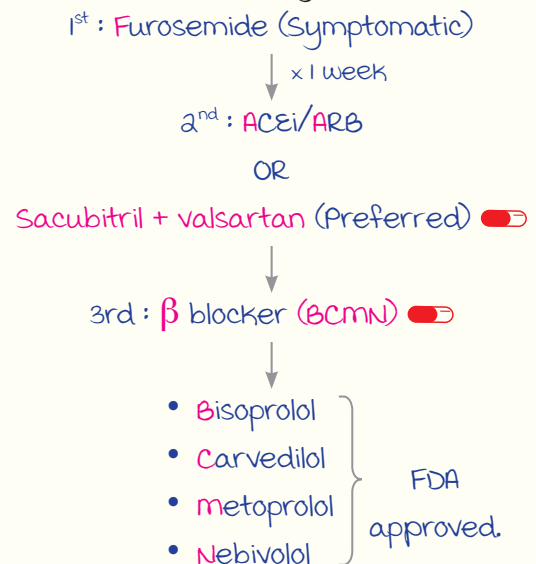
mx :

**Chronic CHF :**

Drugs :

- For symptomatic mx :
 - Furosemide.
 - Digoxin.
- To ↓ mortality :
 - SHIVA Beta** (↓ Remodelling) :
 - **Sacubitril** (⊖ NEP).
 - **Spironolactone**.
 - **SGLT-2** (⊖) (-gliflozins).
 - **Hydralazine** + IDN
 - **Ivabradine**.
 - **Vericiguat** (↑ cGMP).
 - **ACEi/ARB**.
 - **β blocker** (FDA approved).

mx of new case orally :



Hypertension

00:31:00

----- Active space -----

mild to moderate HTN :

	< 55 yr (Young)	≥ 55 yr (Old)
Pathology	↑ Renin	↓ Renin
Drugs	• DOC : ACEi/ARB • Beta blocker	• DOC : CCB (Amlodipine) • Diuretics (Thiazides)

Resistant hypertension : Resistant to at least 3 classes of drugs with one being a diuretic (Pathology : ↑ Aldosterone).

↓
Add

Spironolactone (DOC) .

New drug : Aprocitan (Endothelin receptor .

Drugs for HTN with co-morbidities (Irrespective of age) :

- DM
 - CKD (No edema)
 - Nephrotic syndrome
 - Scleroderma
 - Osteoporosis
 - Edema
 - Raynaud's disease
 - Cyclosporine-induced HTN
 - BPH
 - migraine
 - Hyperthyroidism
 - Stable angina
 - Anxiety disorder
 - Essential tremor
- ACEi/ARB.
- Thiazides (↑ Ca²⁺ absorption from tubules).
- CCBs.
- α : Prazosin.
- Beta blocker.

Severe Hypertension :

	Hypertensive emergency	Hypertensive urgency
BP	↑	≥ 180/110 mmHg
End organ damage		
Drugs	IV drugs : • Labetalol : (DOC pregnancy) • Nicardipine (DOC)	Oral drugs : • DOC : Clonidine • ACEi/ARB • CCB

----- Active space -----

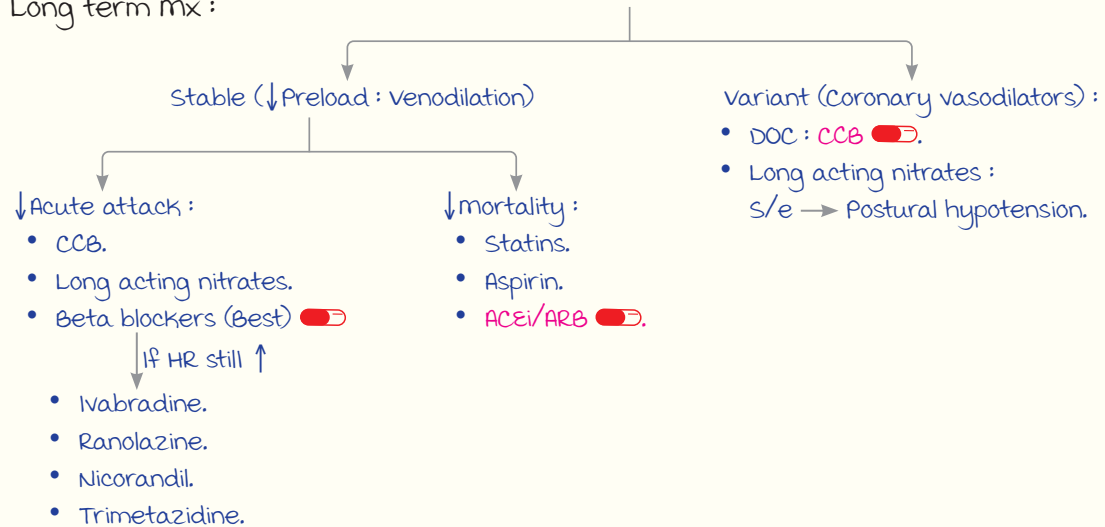
Anti-anginal Drugs

00:38:40

Stable & Variant Angina :

Acute angina Rx : S/L NTG (DOC).

Long term mx :



MOA :

Drug	MOA
Ivabradine	⊖ Funny channels/IF current → ⊖ SA node : ↓ HR, normal contraction
Ranolazine	⊖ Late inward Na ⁺ channel
Nicorandil	K channel opener (Vasodilator)
Fasudil	⊖ Rho Kinase (Vasodilator)
Trimetazidine (Antimetabolite)	⊖ Partial FA oxidase (P-fox) → ⊖ TG metabolism

Side Effects

00:47:04

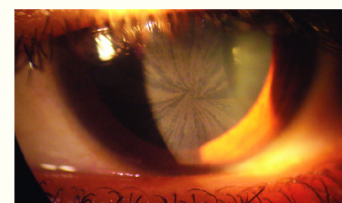
Amiodarone :

Mnemonic : Potassium channel blocker makes liver, nerve and skin toxic.

- Pulmonary fibrosis : C/i in ILD.
- Corneal microdeposits : Whorl-like.
- Blue colored skin (Ceruleoderma) : Sunscreen.
- Myocarditis.
- Liver toxicity.
- Neuropathy.
- Alpha receptor ⊖ causes hypotension.
- Photosensitivity (Brown skin) : Sunscreen.
- Thyroid (Hypo > hyper).



Ceruleoderma



Corneal microdeposits

Note :

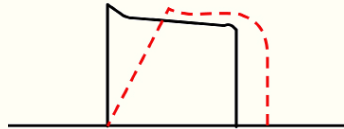
----- Active space -----

Amiodarone :

- Longest acting anti-arrhythmic.
- Not nephrotoxic.

Anti-arrhythmics Causing QT Prolongation/Torsades :Class Ia :

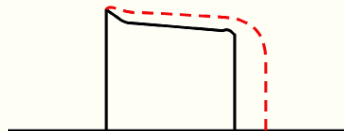
- Quinidine.
- Procainamide.
- Disopyramide.



- Na^+ & K^+ channel $\ominus$.
- Delay in repolarisation & depolarisation.

Class III :

- Black : Bretylium
- D : Dofetilide.
- I : Ibutilide.
- V : Vernakalant.
- A : Amiodarone.
- S : Sotalol.



- Only K^+ channel $\ominus$.
- Delay in repolarisation.

Adenosine :s/e :

- m/c : Flushing and dyspnea (Short acting $\rightarrow$ Transient).
- most important : Atrial fibrillation.

Contraindications :

- Bronchial asthma/COPD.
- Transplanted heart (No vagus $\rightarrow$ Complete heart block).

Digoxin :MOA : Blocks Na^+-K^+ ATPase.s/e : mnemonic DIGOXIN.

- Dialysis, defibrillation : C/i.
- Increases K^+ .
- Gynaecomastia.
- Ocular s/e : Green halos.
- Xanthopsia : Yellow vision.
- Increases risk of arrhythmia :
 - m/c : v. bigeminy.
 - Not seen : Atrial flutter, mobitz type II.
- Nausea/vomiting : m/c & earliest.

C/i :

- Increased risk of toxicity.
mnemonic : KMC in Manipal Rocks
 - $\downarrow \text{K}^+$.
 - $\downarrow \text{Mg}^{2+}$.
 - $\uparrow \text{Ca}^{2+}$ (Increase).
 - MI
 - Renal failure.
- WPW syndrome :
AV node $\ominus \rightarrow \uparrow$ WPW pathway.
- HOCM : $\uparrow$ Contraction.

----- Active space -----

Digoxin toxicity :

mild arrhythmia : v. bigeminy

Rx : K^+ .

Severe arrhythmia : VT, v. fib

- Digibind : Antidote.
- Lidocaine.
- K^+ : c/i.

RAAS Inhibitors :ACE inhibitors :

- 'Pril' : Oral.
- Enalaprilat : IV.
- s/e :
 - Dry cough (m/c).
 - Angioedema ($\uparrow$ Bradykinin) $\rightarrow$ Tracheostomy.

Contraindications :

- Pregnancy.
- B/L renal artery stenosis.

DRI : Aliskiren (Renin $\ominus$).ARBs :

- Losartan  :
 - PPAR- γ $\oplus$: $\downarrow$ Insulin resistance.
 - $\uparrow$ uric acid excretion : DOC HTN + gout.
 - Thromboxane A_a $\ominus$: Anti-aggregation.
- Telmisartan  :
 - max. PPAR- γ $\oplus$: $\downarrow\downarrow$ Insulin resistance.
 - DOC : DM + HTN

Calcium Channel Blockers :

- Headache.
- Ankle edema



Prevention : + ACEi/ARB

- Constipation : verapamil.
- AV block : Diltiazem/verapamil.



c/i with beta blockers

(Causes total conduction block $\rightarrow$ Asystole).Other Drugs :

	s/e	uses
Minoxidil	Hirsutism	Topical use : Androgenic alopecia
Diazoxide	Severe hypotension	DOC : Insulinoma 
Nitroprusside	Cyanide toxicity	Hypertensive emergency

Note :

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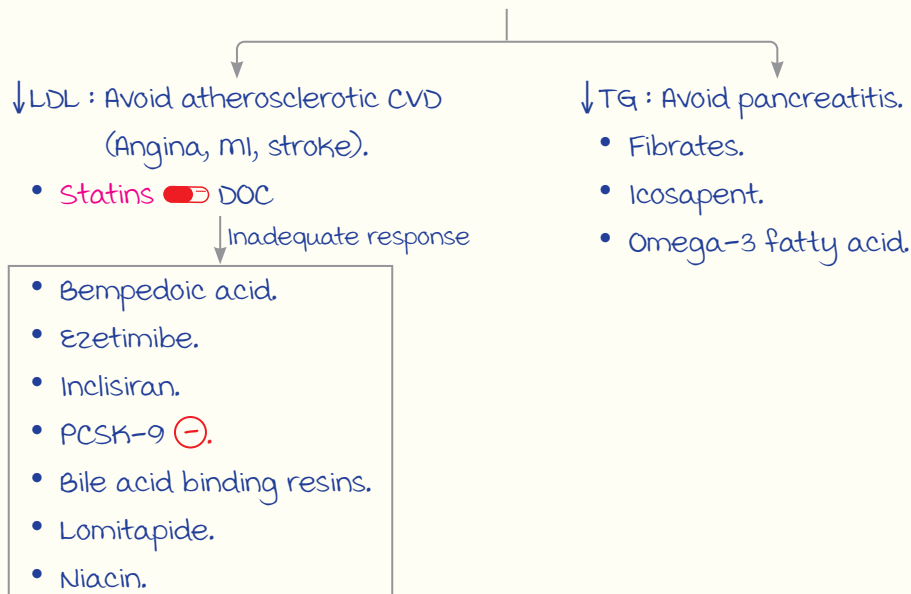
Finasteride :

- DOC Androgenic alopecia : Oral 1 mg .
- BPH : 5 mg.

Hypolipidemic Drugs

01:03:10

Aim :



Statins :

- most potent : Pitavastatin > Rosuvastatin 
- ↓

Ceiling effect

↓

DOC (max. ↓ LDL).
- Longest acting : Rosuvastatin > Atorvastatin.
 - All metabolized by CYP450 except : Pravastatin 
- ↓
- DOC : Protease inhibitors causing dyslipidemia (Enzyme  → Statin toxicity).

MOA :

- Hypolipidemic effect ( HMG CoA reductase) :
 - ↓ LDL, VLDL, TG.
 - ↑ HDL, Lipoprotein-A.
- Pleiotropic effects (All except ↓ LDL) :
 - Antiaggregant, anticoagulant.
 - Anti-inflammatory, antioxidant.
 - ↑ NO.
 - Plaque stabilization.

----- Active space -----

Uses :

- **DOC**  :
Type II hyperlipoproteinemia
(Familial hypercholesterolemia).
- Primary & secondary prophylaxis
of MI and stroke (ASCVD).
- **Night-time dosing**, except :
 - Rosuvastatin } very long
 - Atorvastatin } acting.

s/e :

- **myopathy**.
- Hepatotoxicity.
- Insulin resistance.

c/i :

- Pregnancy.
- Children : < 10 yrs.
- Gemfibrozil :  elimination
(Liver : OATP-1B1 pump).

Drugs	MOA	uses	s/e
PCSK-9   : • Evolocumab • Alirocumab	Prevent LDLr degradation : • ↓ LDL (max) • ↓ Lipoprotein-a	Add on to statins	-
↓ PCSK-9 synthesis : Inclisiran	Is a small interfering RNA → Breaks PCSK-9 mRNA : ↓ LDL		
Evinacumab	Blocks angiotensin-like protein 3 & LPL : ↓ LDL, TG	Familial hypercholesterolemia	-
Lomitapide	Blocks MTP (microsomal triglyceride transport protein)		
Fibrates  : • Clofibrate • Fenofibrate • Bezafibrate • Gemfibrozil	 PPAR-α , ↑ LPL synthesis : • ↓ TG, chylomicrons • ↓ VLDL	DOC  : • Hypertriglyceridemia • Chylomicronemia syndrome • Type III hyperlipoproteinemia	• Cholelithiasis • myopathy
Icosapent	• ↓ TG rich VLDL synthesis by liver •  Platelet aggregation → ↓ CVS mortality	• Hypertriglyceridemia • Add on to statins	-
Niacin (Not used)	• max. ↑ HDL • ↓ Hormone sensitive lipase synthesis • ↓ Lipoprotein-A	Dyslipidemia with ↓ HDL	Toxicity : • Hepatotoxicity • Insulin resistance • Flushing (D/t PG) ↳ DOC : Aspirin
Bempedoic acid	Competitive  ATP citrate lyase : • ↓ Cholesterol synthesis • ↓ LDL	Add on to statins	-
Bile acid-binding salts : • Cholestyramine • Colestipol • Colesevelam	 Enterohaptic circulation of bile acid : ↓ LDL	• Add on to statins • Preferred : - Pregnancy - Children	• Hypertriglyceridemia • Hyperchloremic acidosis • GI upset • ↓ Absorption of other drugs and vit. A, D, E, K • Least with Colesevelam

Drugs	MOA	Uses	s/e
Ezetimibe	$\ominus$ NPC ₁ L ₁ receptor (Small intestine) $\downarrow$ $\ominus$ Cholesterol absorption	Add-on to statins	-

----- Active space -----

Note :

Lipoprotein-A :

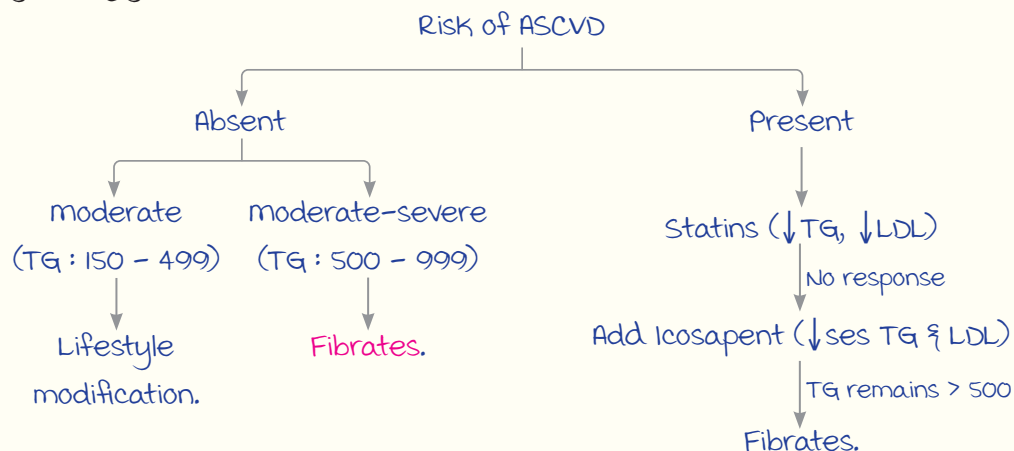
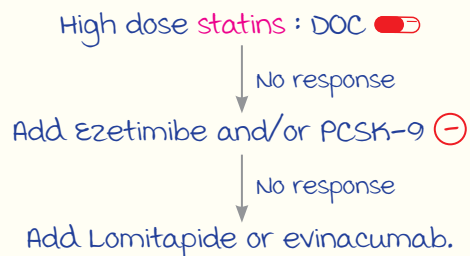
- $\uparrow$: Statins.
- $\downarrow$: Niacin, PCSK-9 $\ominus$.

management :

High LDL :



Familial hypercholesterolemia :



DRUGS ACTING ON KIDNEY

Diuretics

00:00:08

	MOA & site	uses	Side effects
Carbonic anhydrase inhibitors	⊖ Carbonic anhydrase in PCT	- Acetazolamide : DOC  for acute mountain sickness - Acetazolamide/dichlorphenamide : Familial hypokalemic periodic paralysis	↓K⁺ - metabolic acidosis - Renal stones - Hypersensitivity : Rash, BM suppression ↑NH₃ : C/i in cirrhosis
Loop diuretics	⊖ Na/K/ATP pump in thick ascending limb	- Furosemide : DOC  for pulmonary edema - Torsemide : Longest acting - Bumetanide : most potent - Ethacrynic acid : most ototoxic - most effective drugs when GFR < 40	↓K⁺ ↓Ca²⁺ : Rx of hypercalcemia - Ototoxicity (C/i : Aminoglycosides)
Thiazides	⊖ Na/Cl cotransporter in DCT	- DOC for :  - HTN with edema - Nephrogenic DI - Other edema (As add-on) - Chlorthalidone : Longest acting, preferred in HTN - Indapamide : Predominantly hepatic excretion, used in HTN - metolazone : Effective in GFR < 40 (Add-on to furosemide)	↓K⁺ (max) ↑S. Ca²⁺ / ↓U. Ca²⁺ : Rx of hypercalciuric renal stones (+ Citrate to avoid stones) - Thiamine deficiency
Potassium sparing diuretics	- Spironolactone / eplerenone : ⊖ Aldosterone - Amiloride : ⊖ ENaC in collecting duct	Spironolactone/eplerenone : - DOC for :  - Cirrhotic edema (+ Loop diuretic) - Resistant HTN Amiloride : - DOC for :  - Lithium induced DI - Liddle syndrome - used in : Cystic fibrosis	↑K⁺ - metabolic acidosis - Spironolactone : Gynaecomastia
Osmotic diuretics	Solute free water loss : PCT & loop	mannitol (mnemonic : ABCDE) : - Acute congestive glaucoma (DOC ) - Braking of diuretics (Resistant) - Cerebral edema (DOC ) - Dialysis disequilibrium (DOC ) - Expected renal failure	↓/↑Na⁺ ↓/↑K⁺ - Pulmonary edema - C/i in renal failure

Vasopressin Related Drugs

00:14:41

----- Active space -----

Drug	uses	Side effects
Vasopressin antagonist :		
• Conivaptan (iv) • Tolvaptan • mozavaptan } Oral (iv → Emergency; Oral → Long term Rx)	SIADH : 3 rd line → Vasopressin antagonists TOC : • Free H₂O restriction : 1st line • Saline infusion : 2nd line	• Hypokalemia • Hepatotoxicity (Tolvaptan)
Vasopressin analogs :		
Vasopressin	-	-
Terlipressin 	DOC in acute variceal bleed	-
Desmopressin 	DOC in : • Central DI • Nocturnal enuresis • VW disease • Hemophilia A	-

.....
 Note :

Terlipressin > octreotide for Rx of acute variceal bleed as it ↓ mortality.

DRUGS ACTING ON CNS

Anti-Epileptic Drugs

00:00:14

Epilepsy	DOC 
- GTCS - myoclonic seizure	Valproate : $\ominus$ Ca^{2+} & Na^+ channels $\uparrow$ GABA
Absence seizure	Ethosuximide > valproate
- Partial seizure - Rolandic epilepsy	Carbamazepine
mixed seizure syndrome : - Lennox Gastaut syndrome (LGS) - Dravet syndrome (DS)	valproate (Standard drugs : Ineffective)
Seizure in neonates	Phenobarbital (Protective against ischemia)
Alcohol withdrawal seizures	- Diazepam - Lorazepam (Safe in liver failure)
Status epilepticus	1. Lorazepam 2. In children (with no IV access) : Intranasal midazolam
Febrile seizures	Intranasal midazolam >> Rectal diazepam
Infantile spasm with tuberous sclerosis (TS)	Vigabatrin ($\ominus$ GABA transaminase)
Infantile spasm without TS/ Salaam spasm/West syndrome	ACTH

.....
 Note :

valproate has no action on glutamate.

Newer Drugs :

Cannabidiol : uses $\rightarrow$ LGS, TS, DS.

Rufinamide : use $\rightarrow$ LGS (Add-on drug).

Lacosamide : MOA $\rightarrow$ Slow inactivation of Na^+ channels.

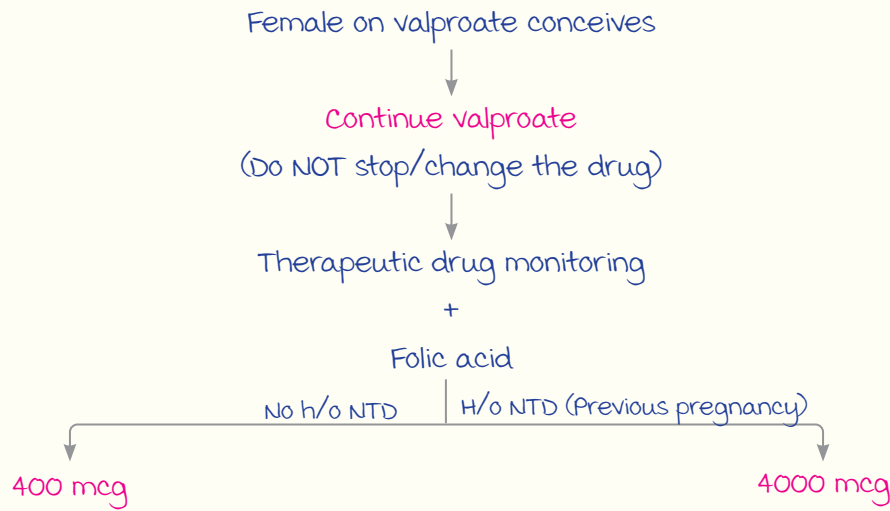
In Pregnancy :

----- Active space -----

Teratogenicity :

- min : **Levetiracetam** (DOC)  < Lamotrigine.
- max : **Valproate**
 - A/w neural tube defects.
 - Category D drug.

mx :

Psychiatric Drugs

00:11:15

mania :

mania		DOC 
Acute mania		Atypical antipsychotics (Aripiprazole/Brexipiprazole)
mania in pregnancy	1 st trimester	Aripiprazole
	2 nd /3 rd trimester	Lithium
• mania prophylaxis • Hypnic headache		Lithium
Bipolar disorder		Lithium (Can ↑ or ↓ BDNF)
Rapid cyler : BPAD with > 4 episodes of mania/depression in 1 yr		Valproate

Teratogenicity in BPAD :

- min : **Lamotrigine**.
- max : **Valproate** >> Lithium.

----- Active space -----

Depressive Disorders :

Depression	DOC	Features
Depression	1 st line : SSRI/SNRI • Early morning dose • Add benzodiazepine (≥ 1 month) To prevent anxiety/insomnia	Withdrawal : • Not seen with : - SSRI : Fluoxetine (Longest acting : 10 days) • Seen with : - SSRI : Paroxetine (Shortest acting) - SNRI : Venlafaxine
With insomnia or erectile dysfunction	mirtazapine : • NaSSA : Noradrenergic & specific serotonergic antidepressants • Night dose (Sedation)	$\ominus H_1$ & $\ominus 5HT_a$
With suicidal tendency	mnemonic : Life Can End. • Lithium • Clozapine • ECT (Best)	-

Novel antidepressants :

- **Vilazodone.**
 - **Vortioxetine** (mSAA : multiple serotonin agonist & antagonist).
 - **Ketamine** (Intranasal).
 - **Zuranolone** (Oral)
 - **Brexanolone** (iv)
- } DOC for postpartum depression .

Degenerative Disorders

00:20:24

Parkinson's Disease :

Effect of disease on lifestyle	mx	s/e
minimal	1. No drug.	-
	Preference 2. MAO-B $\ominus$ (DOC) : a. Selegiline/Rasagiline b. TOC in young onset Parkinson's	-
	↓ 3. NMDA $\ominus$: Amantadine	• Ankle edema • Livedo reticularis
Significant	Age < 65 : D _a agonist (DOC) • Pramipexole. • Ropinirole. • Rotigotine : Transcranial patch	• Fatigue • Compulsive → Gambling → Sexual activity
	Age ≥ 65 : Levodopa • Used with carbidopa : $\ominus$ Peripheral DOPA decarboxylase • On-off phenomenon seen : - D/t fluctuating dopamine - Rx : • COMT $\ominus$: Entacapone • MAO-B $\ominus$: Safinamide • D_a agonists	• Dyskinesia - Rx DOC : Amantadine • Orthostatic hypotension • Psychosis : - Rx DOC : Pimavanserin - Also an absolute C/i • Angle closure glaucoma : - Also a relative C/i

Note :

Drugs causing ankle edema : Amantadine, Amlodipine.

Amyotrophic Lateral Sclerosis :

Aim of mx :

1. Inhibit disease progression.
2. Reverse muscle spasticity.



Livedo reticularis

----- Active space -----

Drugs : (REST + Bed)

- Riluzole : DOC
 - Edaravone : 2nd line
 - Sodium phenylbutyrate taurursodiol (SPT) : 2nd line
 - Tofersen (Only in SOD-1 gene mutation)
 - Baclofen (GABA-B) → Neuro degeneration + muscle relaxant.
- } Neuro degeneration

Alzheimer's Disease :

mx :

- AChE : DOC
 - Donepezil (Oral).
 - Rivastigmine (Oral/patch).
- NMDA : memantine
 - Add-on drug.
 - used in : moderate - severe alzheimer's.
- New drugs :
 - Aducanumab.
 - Lecanemab.
 - Donanemab.

Miscellaneous

00:35:48

Disorders	DOC	MOA/Side effects
Trigeminal neuralgia	Carbamazepine	-
• Postherpetic neuralgia • Pain d/t spinal cord injury • Restless leg syndrome	Pregabalin/Gabapentin	$\alpha_2\delta_1$ subunit of presynaptic voltage-gated Ca^{2+} channel
Psychosis	• Weight neutral atypical antipsychotics : - Aripiprazole - Brexpiprazole - Cariprazine - Ziprasidone	Partial agonist at D_2 & $5HT_2$
	• Last line : - Olanzapine - Clozapine (DOC for resistant psychosis)	-
ADHD	methylphenidate	• Worsens tics • ↑ Risk of abuse

----- Active space -----

Disorders	DOC	MOA/Side effects
ADHD with Tourette syndrome	Clonidine	Withdrawal HTN (D/t α_a downregulation)
ADHD with family h/o drug abuse	Atomoxetine	Selective norepinephrine reuptake $\ominus$
- Tics a/w Tourette syndrome - Huntington's chorea	Tetrabenazine	-

Migraine

00:41:13

Prophylaxis :

- Propranolol (DOC) .
- Topiramate (Toxic).

Treatment :

1. 5-HT_{1B/1D} $\oplus$:

- DOC
 - Oral : Rizatriptan (Fastest oral drug) > Sumatriptan.
 - S/C : Sumatriptan (Overall fastest).
- C/i : Angina (D/t vasoconstriction).

2. New drugs :

- 5-HT_{1F} $\oplus$: Lasmiditan (For acute attack).
- CGRP (Calcitonin gene-related peptide) $\ominus$:
 - Eptinezumab
 - Fremanezumab
 - Galcanezumab
 } S/C, prophylaxis.
- CGRP receptor $\ominus$:
 - Erenumab
 - Olcegepant
 - Rimegepant
 } Oral for either Rx/ prophylaxis.

3. Ergotamines :

Ergotism : Very potent vasoconstrictor → Gangrene of organs with end arteries.

Sleep disorders

00:47:25

Narcolepsy :

modafinil : DOC

MOA : ↑ Dopamine → ↑ Wakefulness.

Insomnia :

Z-compounds : DOC

- Zolpidem, Zaleplon, Zopiclone.

DORA (Dual orexin receptor antagonist):

- Suvorexant.

- C/i in drug abusers.

- DOC in h/o drug abuse.

Sleep induction and maintenance.

melatonin agonist :

- Ramelteon
 - Less effective
 - Less dependence
- } Only sleep induction.

----- Active space -----

Opioids

00:49:52

Condition		Drug
Opioid dependence		↓ withdrawal symptoms : • methadone • Buprenorphine Prevent relapse : Naltrexone (Opioid receptor antagonist)
Labour analgesia		• DOC  : morphine (Other uses : MI, cancer) • Pethidine
Opioid induced constipation		 methyl naltrexone : DOC
Post-op ileus		 Alvimopan : DOC
Post anesthetic chills		•  Pethidine • Tramadol
Opioid contraindicated in MI		Pentazocine
Opioid toxicity		Naloxone (IV → Emergency; I/N → Ambulatory)
Opioid with :	Partial agonist at μ & antagonist at κ	Buprenorphine
	Partial agonist at μ & full agonist at κ	Pentazocine
	5-HT & NE reuptake 	Tramadol
	MAO  effect	Pethidine
	QT prolongation	methadone
Rett syndrome		Trofinetide
Friedreich's ataxia		Omaveloxolone

Note :

- Constipation
 - Convulsion
 - Constriction of pupil (miosis)
- } No tolerance for these side effects.

Mx of Dependencies

00:55:50

Alcohol Dependence :

FDA approved :

1. Aversive Rx : Disulfiram.

- MOA :  Aldehyde dehydrogenase → ↑ Acetaldehyde (Toxic).
- Prerequisites :
 - motivated patient.
 - Abstinence of 12 hours.

----- Active space -----

2. Anti-craving drugs :

Naltrexone (DOC 

$\ominus$ μ receptor $\rightarrow$ $\downarrow$ Euphoric effect.

Acamprosate.

Non-FDA approved :

- Benzodiazepines.
- Clonidine.
- Topiramate.
- Baclofen.

Smoking Dependence :

First line Rx :

1. Drugs :

- Bupropion : Anti-depressant.
- Varenicline - Partial agonist at nicotinic receptor ($\alpha_4\beta_2$).
- most effective.

Second line Rx :

- TCA.
- Clonidine.
- Cytisine.

2. Nicotine replacement :

- Nasal spray :
 - Needs prescription (Schedule H).
 - most effective.
- Patch.
- Gums.
- Lozenge.

Side Effects of CNS Drugs

00:59:34

Anti-Epileptic Drugs :

valproate :

mnemonic : Tab VALPROIC.

- Tab : Tremor, teratogenic (max).
- Vomiting & nausea.
- Alopecia.
- Liver toxicity (C/i in children < 2 years).
- Pancreatitis.
- Rash.
- Obesity : PCOS.
- Increase ammonia.
- Carnitine : Antidote for hepatotoxicity & hyperammonemia.

Phenytoin :

mnemonic HYDANTOIN :

- Hirsutism.
- Hyperplasia of gum.
- Lymphadenopathy.
- Diplopia.
- $\downarrow$ vit D (Hypocalcemia).
- Ataxia $\rightarrow$ Therapeutic drug monitoring & adjust dosage.
- Nystagmus.
- Teratogenic : Facial clefts.
- Osteomalacia.
- Increased bleeding in Newborn ($\downarrow$ vit K).
- Neutropenia, megaloblastic anemia.

Pharmacology • World of Revision • v1.0 • Marrow • 2025

Carbamazepine : Alternative → Oxcarbazepine/Eslicarbazepine :

----- Active space -----

Mnemonic : **HEADS**.

- Hyponatremia (D/t SIADH) :
 - Delayed s/e.
 - m/c in elderly.
- Hypersensitivity :
 - Eosinophilia.
 - Aplastic anemia.
 - Agranulocytosis.
- Ataxia.
- Diplopia.
- SJS : HLA-B*57:02 gene.

- ↓ Hypersensitivity.
- ↑ Hyponatremia.

Lamotrigine (No action on GABA) :

SJS (Rash) : Started at low dose.

Topiramate :

- mnemonic : **RAM ate less**.
 - Renal stones.
 - Angle closure glaucoma.
 - metabolic acidosis.
- Weight loss.
- use : Obesity.

Anti-Depressants :

Drugs		s/e	
		Brain	Spinal cord
SSRI/ SNRI	5-HT _a ⊕	- Anxiety - Insomnia - Vivid dreams	- Erectile dysfunction - Delayed ejaculation - Anorgasmia
	5-HT ₃ ⊕	Nausea, vomiting	
	5-HT ₄ ⊕	Loose stools	
TCA	muscarinic ⊖	- Constipation, dry mouth - Urine retention, mydriasis	c/i in glaucoma, BPH
	5-HT, H ₁ ⊖	Obesity, sedation	
	α ₁ ⊖	Postural hypotension	
mirtazapine H ₁ ⊖		Sedation, obesity	
Bupropion		Seizure, max anxiety	
MAO ⊖		- Cheese reaction - Serotonin syndrome	Rx Phentolamine Lorazepam → Cyproheptadine (DOC)
Trazodone : 5-HT _a ⊖		Priapism	

→ 2° use : Premature ejaculation (Dapoxetine)

Toxicity :

- DOC : Bicarbonate.
- Cause of death: Arrhythmia.

----- Active space -----

Antipsychotics :Affinity towards a receptor $\propto$ s/e at that receptor :

Drugs		Side effects	Drugs with $\uparrow$ affinity & $\uparrow$ s/e
D_a receptor $\ominus$		• EPS • Hyperprolactinemia	Atypical : Risperidone Typical : Haloperidol
Other receptor $\ominus$:	muscarinic $\ominus$	• Constipation • Dry mouth • mydriasis, urine retention	Clozapine > Olanzapine
	H_1 $\ominus$	Sedation	
	H_1 & 5-HT $\ominus$	Obesity	
	α_1 $\ominus$	• Postural hypotension • metabolic s/e : Dyslipidemia & hyperglycemia	

Zero Side Effects Seen With :

- New drug : **Xanomeline** ($\oplus$ m_1 & m_4 , no action on D_a)
- Others :
mnemonic : **ABCZ**.
 - **A**ripiprazole.
 - **B**rexipiprazole.
 - **C**ariprazine.
 - **Z**iprasidone.

Extrapyramidal Symptoms

01:10:35

EPS	Symptom	Cause	DOC
Akathisia : m/c	Restlessness	Unknown	• Beta blockers $\ominus$: DOC • Benzodiazepines
Acute dystonia (Earliest)	• Abnormal posturing • Facial grimacing	• D_a $\ominus$ • metoclopramide inj.	Anticholinergics : DOC $\ominus$ • Trihexyphenidyl (Benzhexol) • Benztropine • Biperiden • Antihistaminics : Promethazine
Parkinsonism	• Tremor • Bradykinesia		
Tardive dyskinesia (most late)	• Facial dyskinesia : - Tongue protrusion - Lip smacking • Limb dyskinesia : - Piano finger movement - Foot tapping	D_a upregulation	VMAT-2 $\ominus$: • valbenazine • Deutetrabenazine
Neuroleptic malignant syndrome (most lethal)	• muscle rigidity • Hyperthermia • ANS instability	D_a $\ominus$	• Dantrolene $\ominus$ DOC $\rightarrow$ $\ominus$ Ryr • Bromocriptine $\rightarrow$ $\oplus$ D_a (most specific drug)

Lithium

01:13:19

----- Active space -----

Side Effects :

Mnemonic : **HOTHEAD**.

- Hypothyroidism ($\ominus$ TSH-R).
- Obesity.
- Tremor :
 - Fine : Normal plasma conc.
 - Coarse : Toxicity
- Hypercalcemia ($\uparrow$ PTH).
- Ebstein anomaly :
c/i in pregnancy (1st trimester).
- Acne.
- Diabetes insipidus (Thirst) :
Rx DOC $\rightarrow$ Amiloride.

Interactions :

- Thiazide $>$ K^+ sparing diuretic.
 - Vomiting, diarrhea.
 - Fasting.
 - RAAS ($\ominus$)
 - NSAIDs
- } $\downarrow$ s. Na $\rightarrow$ $\uparrow$ Li tubular reabsorption.
- } $\downarrow$ Filtration of Li.
- mannitol $\rightarrow$ $\uparrow$ excretion of Li.
 - Li + NDMR $\rightarrow$ $\uparrow$ $\uparrow$ Effect of NDMR (But Li not stopped pre-op).

Therapeutic Drug monitoring :

- Sample taken 12 hrs after last dose + 6th day of Rx (Half life : 24 hrs).
- Doses (mEq/L) :
 - Prophylaxis of mania : 0.6 - 1.
 - Acute mania : 1 - 1.5.
 - Toxicity : $>$ 1.5.
 - $>$ 4 $\rightarrow$ Dialysis (TOC).

Side Effects of Opioids

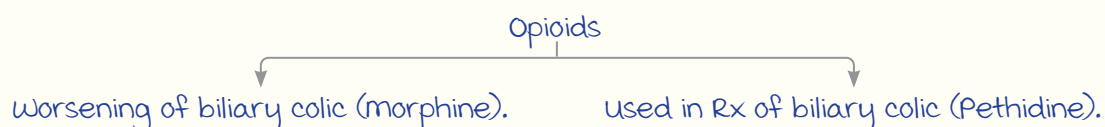
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μ Receptor Effects :

Mnemonic : **MUSCARINE**.

- miosis.
 - urine retention.
 - Sedation.
 - Constipation, convulsion.
 - No bile flow d/t contraction of sphincter of Oddi.
 - Euphoria.
- } Opposite effects
withdrawal symptoms. {
- Analgesia.
 - Respiratory depression.
 - Increase muscle rigidity.

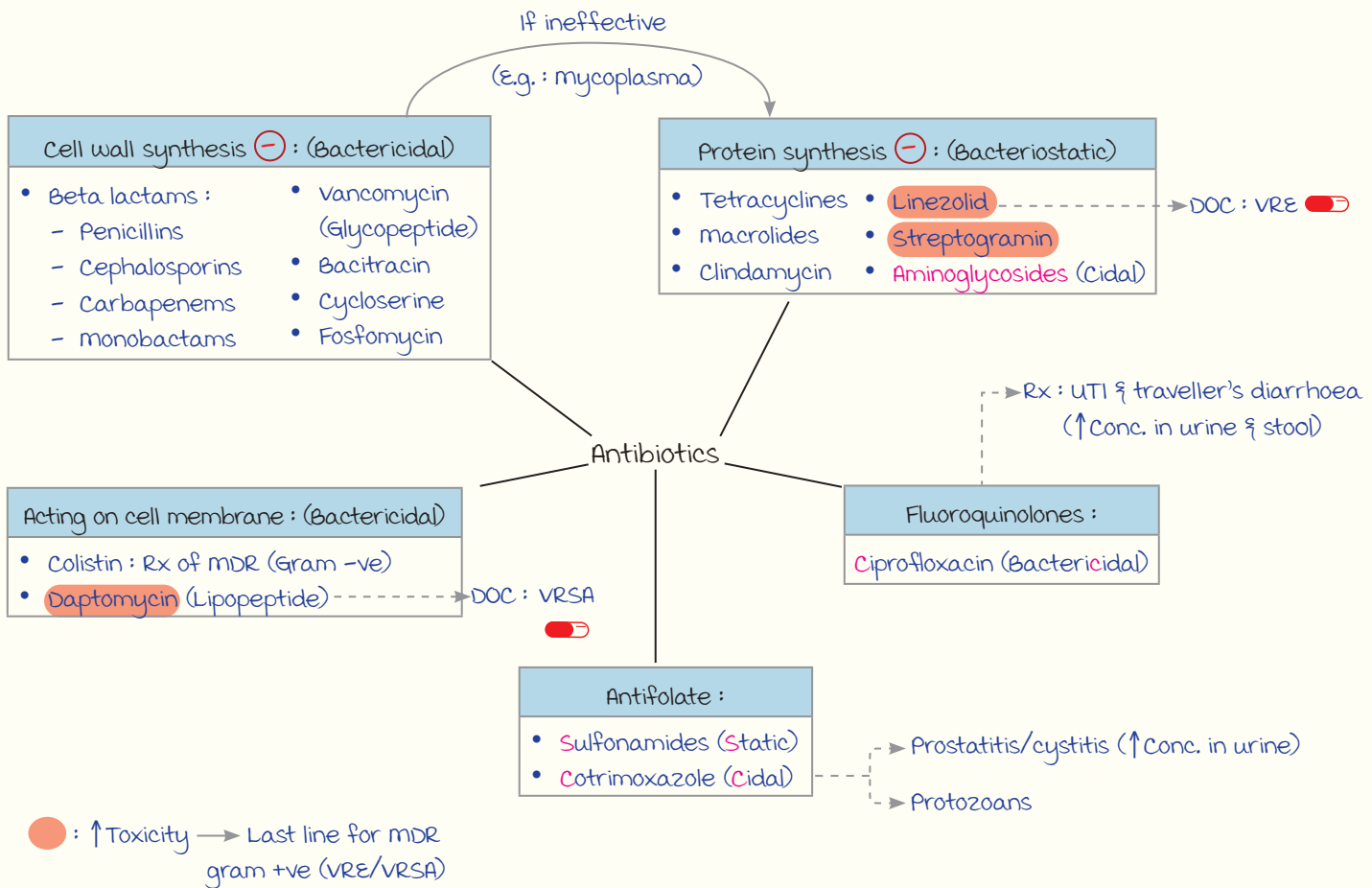
Note :



ANTIMICROBIAL DRUGS : PART 1

Classification

00:00:17



Regimens for some severe infections :

- 1st line : β -lactam (Least toxic) → 2nd line : Vancomycin $\xrightarrow{\text{Severe infection}}$ Add on : Aminoglycoside.

E.g. :

- Pseudomonas infection $\xrightarrow{\text{mild to moderate infection}}$ Ceftazidime $\xrightarrow{\text{Severe infection}}$ Ceftazidime + Gentamicin/ Aminoglycoside.
- Endocarditis → Ampicillin $\xrightarrow{\text{Resistance}}$ Vancomycin + Gentamicin.

Cell Wall Synthesis Inhibitors

00:05:58

----- Active space -----

1. Fosfomycin :



2. Cycloserine :



3. Bacitracin :

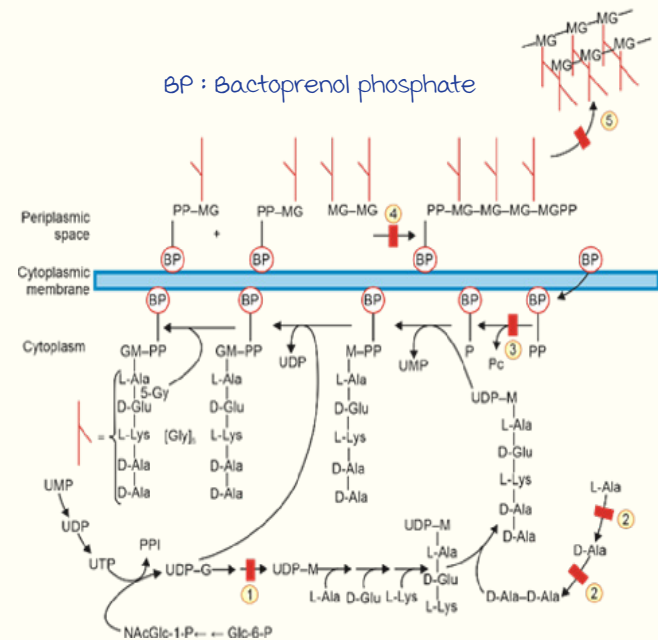
⊖ Bactoprenol phosphorylation.

4. Vancomycin :

⊖ Polymerization.

5. Beta-lactams :

⊖ Crosslinking by blocking transpeptidase/PBP.



Cell wall synthesis

β Lactams :

Penicillins :

Drug	Use
Benzathine penicillin G :	DOC : mnemonic SLY GRAM +ve
• Im (m/c used)	• Streptococcus
• Longest acting (28 days)	• Syphilis
	• Leptospirosis
	• Yaws
	• Gas gangrene
	• Rat bite fever
	• Actinomycosis
	• Meningococcus
Amoxicillin (Oral)	• Respiratory infection
	• UTI (Safe in pregnancy)
Ampicillin (Iv)	Listeria meningitis (DOC)
Piperacillin	Pseudomonas

• Penicillin resistance :

- mediated by β-lactamases (Penicillinase) produced by Staph.
- Penicillinase resistant penicillin (PRPs) :

methicillin

⊖ β-lactamase/penicillinase

Counteracted by :
Altered PBP
(mec-A gene)

MRSA

Vancomycin : DOC

⊖ d-alanine

d-alanine → d-lactate

VRSA

 Daptomycin : Ineffective → Linezolid/
DOC Streptogramin

----- Active space ----- Cephalosporins :

Drugs	Uses
1 st gen :	
Cefazolin (IV)	DOC : Surgical prophylaxis
3 rd gen :	
Ceftriaxone (Parenteral)	DOC : • Typhoid • meningitis (empirical) • Gonorrhea
Cefixime	Typhoid (Oral DOC)
Ceftazidime	Pseudomonas (DOC)
Cefoperazone	Pseudomonas
4 th gen :	
• Cefepime • Cefpirome	• Pseudomonas • Enterobacter • ESBL (Extended spectrum β -lactamase)
5 th gen :	
• Ceftobiprole • Ceftaroline	MRSA (Bind to altered PBP)

Carbapenems :

- DOC for :
 - ESBL.
 - Enterobacter.
 - Acinetobacter.
 - Serratia.
- Anti-pseudomonal activity $\oplus$
- Imipenem $\xrightarrow{\text{Renal dihydropeptidase}}$ metabolised.
 - $\uparrow$ Prolongs action $\xrightarrow{\text{Cilastatin } \ominus}$

Protein Synthesis Inhibitors

00:19:25

Tetracyclines :

MOA : Block acceptor site.

MOR (mechanism of resistance) : Drug efflux.

uses of doxycycline : Oral (mild to moderate infection)/IV (Severe infection).

Mnemonic : my Pink RBC.

- mycoplasma (STD).
- Plague prophylaxis.
- Rickettsia : Scrub typhus.
- Borrelia, Brucella.
- Chlamydia (STD), Cholera.

----- Active space -----

C/i of tetracyclines :

- Pregnancy } Bone growth defect.
- Children }
- To be avoided with milk/antacids (TTC chelates Ca^{2+}).

Note :

Tigecycline  :

- Glycylcycline : Derived from minocycline.
- MOA : Blocks acceptor site.
- MOR : Drug efflux.
- Route : IV (Severe infection).

macrolides :

Drugs	uses
Erythromycin 	DOC : • Pertussis • Diphtheria
Azithromycin 	DOC : • Chlamydia • Legionella • mycoplasma • Campylobacter • Chlamydia, cholera in pregnancy • Cholera in children Atypical pneumonia.

Clindamycin :

uses :

- DOC  for toxic shock syndrome (Cidal drugs ↑ toxin release).
- Anaerobic infections.

Note :

Rx of anaerobic infection —→ Supradiaphragmatic : Clindamycin.
 —→ Infradiaphragmatic : metronidazole.

E.g. : Rx of endometritis —→ metronidazole + Gentamicin (Add-on) >
 Clindamycin + Gentamicin.

----- Active space -----

Aminoglycosides :

MOA : misreading of mRNA.

MOR :

1. Enzymatic inactivation (Absent in Amikacin).
2. Altered ribosome structure : Only seen with Streptomycin.

monotherapy not effective against :

- Anaerobes.
- Typhoid.

Uses :

- Gentamicin/Streptomycin (IV/IM) : DOC ➡ for plague & tularemia.
- Oral Neomycin : Gut sterilization in hepatic encephalopathy.

Note :Rifaximin :

- DOC ➡ for gut sterilization in hepatic encephalopathy.
- Pseudomembranous enterocolitis.
- Traveller's diarrhea.

Miscellaneous Antibiotics :

Fidaxomicin :

- MOA : ⊖ RNA polymerase.
- Route : Oral.
- Use : DOC ➡ for pseudomembranous enterocolitis.

Mupirocin-Bacitracin :

- Route : Topical.
- Use : DOC ➡ for staphylococcal nasal carriers.

Fosfomycin :

Rx of UTI → 3 g single dose.

Chloramphenicol :

S/e : Grey baby syndrome (when used for respiratory infections in children).

Anti-Folate Drugs

00:32:43

----- Active space -----

Sulfonamides :

Drugs	uses	
Sulfadiazine + pyrimethamine (Teratogenic)	DOC : Toxoplasmosis	
Sulfadiazine (Topical)	DOC : Burns	
Cotrimoxazole : Trimethoprim + Sulphamethoxazole (1 : 5 ratio)	DOC : mnemonic → (Cautery PINS)	
	- Cystitis - Cyclospora - Cepacia burkholderia - Pneumocystis	- Isospora - Nocardia - Sarcocyst - Stenotrophomonas

Add-on

Ticarcillin

Note :

Toxoplasmosis in pregnancy : Spiramycin.

Fluoroquinolones

00:36:08

MOA : ⊖ DNA gyrase.

Ciprofloxacin :

- m/c used (most active).
- Well concentrated in urine & stool.

Uses :

DOC for :

- Pyelonephritis.
- Typhoid carriers.
- Traveller's diarrhea.
- Shigella.
- Contact of meningococcal meningitis (Previously DOC : Rifampicin).

Respiratory Fluoroquinolones :

mnemonic : Lungs have maximum Germs.

- Levofloxacin : maximum oral bioavailability.
- moxifloxacin.
- Gemifloxacin.

----- Active space -----

moxifloxacin :

- Longest $T_{1/2}$.
- maximum QT prolongation → Highest risk of torsades de pointes.
- maximum risk of seizure.
- Highest mycobacterial effect.
- maximum hepatic excretion :
 - Safe in renal failure.
 - Ineffective for UTI.

Side Effects

00:40:15

Drugs	s/e & c/i
Tetracyclines	• s/e : mnemonic → Tetra PACKET - Photosensitivity - Acute renal failure (except Doxycycline) - Calcium binding (↓ Bone growth) - Kidney : ↑ urine → Diabetes insipidus - Esophagitis - Teeth : Yellow • c/i : Pregnancy, children, antacids/milk
Aminoglycosides	• s/e : - Nephrotoxicity - Neuromuscular toxicity : Reversed by calcium - Ototoxicity (Outer hair cell : Irreversible damage) ↓ Auditory : Amikacin ↓ Vestibular : Streptomycin (+ minocycline) • c/i : Pregnancy
Erythromycin	- s/e : mnemonic → macro SD Card - motilin receptor (+) : Hypertrophic pyloric stenosis - Skeletal muscle weakness - Diarrhea - Cardiac : QT prolongation - Cholestatic jaundice
Fluoroquinolones	- s/e : - Photosensitivity - QT prolongation - Rash - Seizure : D/t displacement of GABA from binding site - Tendinitis/tendon rupture : ↑ Risk in elderly, renal failure, steroid use

----- Active space -----

Drug	s/e & c/i
Linezolid	s/e : - Bone marrow suppression ($\downarrow$ Platelet count) : $\downarrow\downarrow$ with Tedizolid (But $\uparrow T_{1/2a}$) - MAO inhibition $\rightarrow$ Cheese reaction - Mitochondrial toxicity : Optic neuritis, lactic acidosis, myopathy
β lactams	s/e : - Pseudomembranous enterocolitis : 3rd gen cephalosporin > Amoxicillin - Type I hypersensitivity : Rash, anaphylaxis - Type 2 hypersensitivity : Hemolysis - Disulfiram-like reaction - Hypoprothrombinemia - Seizures : Imipenem - Unsafe in renal failure (Except Cefoperazone : But ineffective in UTI) Cefoperazone, moxalactam, Cefamandole, Cefotetan
Sulfonamides	s/e : mnemonic $\rightarrow$ Folic Acid makes RBC - Fetus $\rightarrow$ Kernicterus after birth - Acute intermittent porphyria - Methemoglobinemia - Rash - Bone marrow suppression - Crystalluria
Vancomycin	s/e : - Red man syndrome (D/t histamine) - Ototoxicity - Nephrotoxicity : matzke nomogram $\rightarrow$ Vancomycin dosing

..... Note :

- Drugs causing QT prolongation : Quinidine, moxifloxacin, macrolides.
- vit D deficiency : Not a/w tendon rupture.

ANTIMICROBIAL DRUGS : PART 2

Non-retroviral Drugs

00:00:04

Drugs	uses	Side effects
Oral valacyclovir  > acyclovir	DOC : HSV/VZV	-
Topical acyclovir	Orolabial herpes	-
IV acyclovir 	DOC : HSV encephalitis	Crystalluria → Renal failure
Oral valganciclovir  > ganciclovir	DOC : CMV retinitis	Bone marrow suppression
IV ganciclovir 	DOC : CMV pneumonia	
IV Foscarnet	Resistant herpes	Electrolyte imbalance : ↓K ⁺ , ↓mg, Ca ↑/↓, PO ₄ ↑/↓
Intralesional cidofovir 	DOC : Recurrent laryngeal papillomatosis	-

Disease	DOC	Other drugs
Influenza	• Oseltamivir (DOC) : Neuraminidase ⊖ ↓ Resistant Inhalational zanamivir • Rx : 5 days • Prophylaxis : 7 days	New drug : Baloxavir  • ⊖ RNA polymerase (Endonuclease) • Oral : Rx and prophylaxis • Single dose
Hepatitis B (Chronic active case (↑LFT))	Tenofovir (DOC), entecavir : • No specific duration • Can be used in hepatic decompensation	IFN-α : • Only 48 - 52 weeks • Avoided : more toxic
Hepatitis C	Direct acting drugs (Oral) : • Sofosbuvir : NSSB ⊖ • Velpatasvir : NSSA ⊖ • Paritaprevir : Protease ⊖	• IFN-α : S/c • Ribavirin : Oral
RSV	Inhalational ribavirin	Prophylaxis : • Palivizumab • Nirsevimab

Antiretroviral Drugs

00:09:54

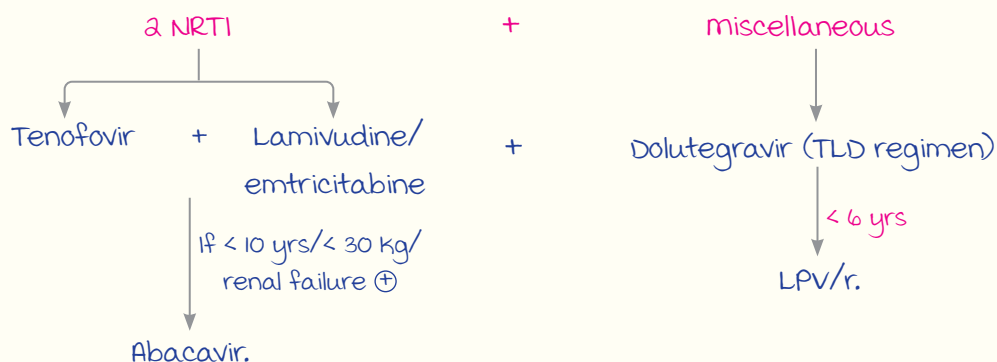
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NRTI	NNRTIs
1. Zidovudine : s/e : Bone marrow suppression 2. Lamivudine $\xrightarrow{\text{Derivatives}}$ Emtricitabine (Hence not used together) : s/e : Pigmentation of palms and soles 3. Tenofovir : • s/e : Nephrotoxic • c/i : - Renal failure - Safety unknown in < 10 yrs, < 30 kg } Abacavir used (s/e : SJS a/w HLAB5701)	Nevirapine DOC : Prevent perinatal HIV transmission in neonates (Alt. : Zidovudine)
Entry inhibitors	Integrase inhibitors
- Ibalizumab (IV) : CD-4 $\ominus$ (HIV I) - Maraviroc : CCR5 $\ominus$ (HIV I & 2) - Fostemsavir : Gp 120 $\ominus$ (HIV I) - Enfuvirtide (S/c) : Gp 41 $\ominus$ (Fusion $\ominus$)	- Dolutegravir - Elvitegravir
Protease inhibitors	
- Used < 6 yrs - Lopinavir : used as LPV/r = 90% LPV + 10% ritonavir - Ritonavir : - most potent enzyme $\ominus$: CYP3A4 $\ominus$ - used as booster : $\ominus$ metabolism - Do not boost : Nelfinavir red pill - Atazanavir red pill : No dyslipidemia - Indinavir : s/e : Renal stones - Tipranavir : s/e : Intracranial bleed	

Note :

Cobicistat : used as a booster with $\rightarrow$ Atazanavir, darunavir, elvitegravir.

Regimen (Rx & post exposure prophylaxis) :



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Antifungal Drugs

00:20:28

Diseases	DOC 	Other drugs
- Kala-azar - mucormycosis - Cryptococcal meningitis	Amphotericin B (IV with 5% dextrose) : S/e : - Hypokalemia : KCl for prevention - Nephrotoxic : - Load patient with 1 - 2 L NaCl - Combined with liposome	-
Candida albicans : mucocutaneous infection (vagina)	Fluconazole (Except oral candidiasis)	vaginal candidiasis (New) : - Oteseconazole (Azole) - Ibrexafungerp (β glucan synthase $\ominus$)
Oral candidiasis	Clotrimazole (Lozenge)	-
Candida : - Non-albicans - Systemic	Echinocandin (β glucan synthase $\ominus$) : - Caspofungin - micafungin - Anidulafungin } IV	-
Aspergillosis	Voriconazole	-
Skin infection	-	Sertaconazole (Topical) : Anti-pruritic
- Tinea cruris - Tinea corporis	Itraconazole	Terbinafine
T. capitis	Griseofulvin : - Orally with fatty food - Dermatophyte infection : Resistance	-
Fungal corneal ulcer	Natamycin	-
Onychomycosis	Terbinafine	- Tavaborole - Efinaconazole } Solution

Note :

Atropine : Used in fungal corneal ulcer $\rightarrow$ Prevents synechiae.

Antihelminthic Drugs

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	DOC : Albendazole	Other drugs DOC
Nematodes	- Roundworm - Whipworm - Hookworm } Soil transmitted - Enterobius vermicularis - Trichinella spiralis	- Strongyloides O. volvulus - Loa loa : DEC - Filariasis : IDA (Ivermectin, DEC, albendazole) - Dracunculus : metronidazole } Ivermectin
Cestodes	- Neurocysticercosis : 1st drug : Prednisolone (↓Edema) - Echinococcus	- Intestinal T. solium T. saginata H. nana D. latum } Praziquantel
Trematodes	-	- Others flukes : Praziquantel - Fasciola hepatica : Triclabendazole

Antiprotozoal Drugs

00:30:22

Disease	mx		
Amoebiasis	Intestinal		Extraintestinal (Alt. : Chloroquine) : DOC : metronidazole
	Asymptomatic	Symptomatic	
	Luminal amoebicide (DIP) : - Diloxanide furoate - Iodoquinol - Paromomycin (DOC)	DOC : metronidazole	
Leishmaniasis	Visceral (Kala azar) : - IV : Liposomal amphotericin B - Oral : miltefosine → DOC : PKDL		Cutaneous : Na ⁺ stibogluconate
Trypanosomiasis	American (Chagas disease) : Benznidazole		African (Sleeping sickness)
			East : - Early : Suramin (IV) - Late : melarsoprol (IV) West : - Early : Pentamidine (IV) - Late : Eflornithine (IV) - Oral : Fexinidazole (DOC)
Babesiosis	- DOC : Atovaquone + azithromycin - Alt. : Quinine + clindamycin		
Toxoplasmosis	DOC : Sulfadiazine + pyrimethamine		
- Cryptosporidiasis - Resistant giardiasis	DOC : Nitazoxanide (PFOR ⊖)		
- Giardiasis - Amoebiasis, anaerobes - Trichomoniasis, tetanus - Bacterial vaginosis	DOC : metronidazole (s/e : Disulfiram like reaction)		

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Antimalarial Drugs :

Severe falciparum malaria :

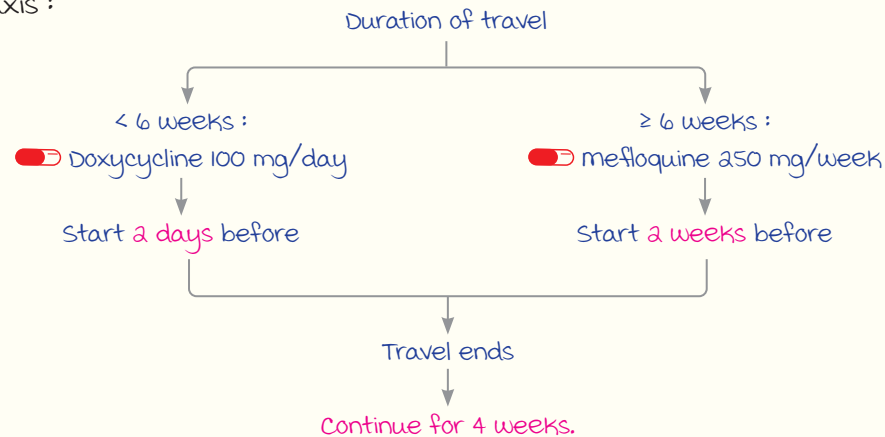
☞ DOC : IV artesunate (48 hrs continuous infusion).

	P. vivax	P. falciparum/resistant P. vivax (ACT)
uncomplicated malaria	☞ DOC : Chloroquine x 3 days + Primaquine x 14 days OR Tafenoquine once	☞ Artesunate + sulfadoxine + pyrimethamine ☞ Artemether + lumefantrine : North-east India - Alt. : Quinine + Tetra/doxy/clindamycin
uncomplicated malaria in pregnancy	Chloroquine	1st trimester : Quinine + clindamycin 2nd/3rd trimester : ACT

Drugs c/i in pregnancy :

- Primaquine.
- Tafenoquine.
- ACT (C/i in 1st trimester).
- Tetracycline.
- Doxycycline.

Prophylaxis :

**Antitubercular Drugs**

00:42:36

First Line :

	Isoniazid (H)	Rifampicin (R)	Pyrazinamide (Z)	Ethambutol (E)
MOA	- Activated by catalase peroxidase ⊖ mycolic acid synthesis	⊖ RNA polymerase	-	-
Action	Cidal	Cidal (max.)	Cidal	Static
Target type bacteria	Replicating	Non-replicating/persisters		Replicating
	Intra + extracellular		Intracellular	Extracellular
Excretion	Liver	Liver (max.) (Safest in RF)	Liver	Kidney (C/i in RF)

Side effects :

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Isoniazid ($\downarrow$ vit. B ₆)	Rifampicin	Pyrazinamide	Ethambutol
$\downarrow$ GABA : Euphoria, hallucination, psychosis $\downarrow$ Heme : Anemia - Rx : vit. B₆ - Toxicity : Seizure - Rx : IV pyridoxine  1 g for 1 g Isoniazid - max 5 g	- Red-orange urine - Respiratory symptoms (Flu-like) - Permanently stop : - Purpura ($\downarrow$ Platelet) - Pulmonary syndrome	- most hepatotoxic : Z > H > R - Hyperuricemia - Hip joint pain (Arthralgia)	- Optic neuritis - Red-green color blindness (Green > red)

New Drugs :

	Bedaquiline	Delamanid	Pretomanid
MOA	⊖ ATP synthase	Nitroimidazole : • Free-radical production • ⊖ mycolic acid synthesis	
About	Bactericidal, taken with food (↑Absorption)		
Side effects	QT prolongation (C/i : Arrhythmia)		-
Pharmacokinetics	• Sequestered in tissues • T_{1/2} : 165 days • Dosing is intermittent : 3 days/week • Safe in pregnancy	• Highly plasma protein bound • Metabolised by albumin • C/i : Albumin < 2.8 mg/dL	metabolised by CYP3A4 : Rifampicin (Enzyme inducer) ↓ses effectivity

Antileprosy Drugs

00:54:26

1st line :

- Rifampicin : most cidal.
 - Dapsone : m/c side effect $\rightarrow$ Haemolysis in G6PD deficiency
 - Clofazimine side effects $\rightarrow$ Skin pigmentation, ichthyosis
- } Static.

2nd line :

- minocycline
 - Clarithromycin
- } Static.
- Fluoroquinolones (Cidal) :
 - moxifloxacin.
 - Oflloxacin.

DRUGS ACTING ON ENDOCRINE SYSTEM

Antidiabetic Drugs

00:00:10

Based on mechanism of action :

↑ Insulin Release :

Sulfonylureas :

- Glyburide
 - Gliclazide
- } s/e : weight gain d/t ↑ adipocytes.

meglitinides :

- Repaglinide.
- Nateglinide.

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Note :
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s/e of hypoglycemia :
Insulin (maximum) >
Sulfonylureas.

GLP-1 agonist :

- Liraglutide (S/C, OD).
- Semaglutide (S/C once weekly/oral OD).
 - DOC for obesity. 🚫
 - s/e : Nausea, vomiting, gastroparesis.

DPP-4 inhibitors (Oral) :

- Sitagliptin.
- Saxagliptin.
- Linagliptin : Hepatic excretion → Safe in renal failure.

↓ Insulin Resistance :

MOA : Stimulation of PPAR-γ .

Drugs : Pioglitazone, Rosiglitazone (Last line).

s/e :

- Hepatotoxic.
- Stimulation of ENaC → Retention of Na^+ & H_2O → Edema, CHF, weight gain.
- Bladder cancer.
- Bone : Fractures in females.
- macular edema.

↓ Hepatic Glucose Production :

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MOA : Activates AMPK.

Drug : metformin.

Uses :

- DOC for Type II DM 🚫.
- Other uses :
 - PCOS : Induces ovulation (Preferred drugs : Letrozole, Clomiphene).
 - MASH (metabolic syndrome associated steatohepatitis).

S/e :

- Lactic acidosis.
- ↓ Vit B₁₂.
- Diarrhoea.

C/i of metformin :

- Elderly
 - Hepatic/Renal failure
 - Chronic alcoholic
 - Severe lung disease
- } ↑ Risk of lactic acidosis.

↑ Urinary Glucose Excretion :

Drugs :

- SGLT-2 inhibitors : Canagliflozin, Dapagliflozin, Empagliflozin.
- SGLT-1 & 2 inhibitor : Sotagliflozin.

Use : Chronic CHF.

S/e :

- D/t ↑ glycosuria : UTI, vaginal candidiasis.
- D/t loss of Na⁺ & H₂O : Hypotension, dehydration.

↓ Intestinal Glucose Absorption :

Drugs : Acarbose, voglibose, miglitol

MOA : α-glucosidase inhibitor (↓ Starch & disaccharide breakdown)

S/e :

- Flatulence.
- Diarrhoea.

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Insulin :

Duration of action	Insulin & features	Side effect
Shortest & fastest acting	- Afrezza (Inhalational) - Taken before food - use : Post prandial hyperglycemia (PPH)	Cough, ↑Risk of lung cancer (C/i in asthma, smokers)
Short acting	- Route : S/C (m/c site : Abdomen) - use : Post-prandial hyperglycemia - Regular : Slow acting (60 min before food) - Glulisine - Aspart - Lispro Fast acting monomeric (15 min before food)	- Hypoglycemia (↑Risk with short acting) - Hypokalemia - Lipodystrophy (Prevention : Rotate site of injection)
Intermediate acting	- Route : S/C - use : maintenance - NPH (Cloudy white) - Lente (30% short acting semilente/powder + 70% long acting ultralente/crystal) - Frequency of dose : BD/TDS	
Long acting	- Route : S/C - use : maintenance - Glargine : white crystals on combining with other insulin - Detemir - Degludec : Longest acting - Frequency of dose : OD/BD	

Amylin Analog :

Drug : Pramlintide.

use : Type I & II DM for PPH (Delays gastric emptying).

S/e : Nausea, vomiting, weight loss.

miscellaneous Drugs :

- Colesevelam
 - Bromocriptine
- use : Type II DM.

updates in Diabetes mellitus :

Tirzepatide :

- MOA : GLP-1 (Glucagon-like peptide) + GIP agonist (Glucose-dependent insulinotropic peptide).
- Route : S/C.
- uses : DM, obesity (more effective than semaglutide).
- S/e : Nausea, vomiting, gastroparesis.

Teduglutide :

- MOA : GLP-2 agonist.
- use : Rx of malabsorption a/w short bowel syndrome.

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Drugs ↓ CV mortality :

1. metformin.
2. GLP-1 agonist.
3. SGLT-2 inhibitors.

↓ Risk of ESRD (End stage renal disease) :

1. GLP-1 agonist.
2. SGLT-2 inhibitor.
3. Finerenone (Aldosterone antagonist).

GnRH Related Drugs

00:16:00

GnRH agonists : Goserelin, Buserelin, Nafarelin & Leuprolide.

GnRH antagonists  : Ganirelix, Abarelix, Cetrorelix & Elagolix.

Dosing, Uses & Side Effects :

GnRH agonists produce different effects when dosed intermittently vs. continuously.

	GnRH agonists (Intermittent)	GnRH agonists (Continuous)/GnRH antagonists	
MOA	↑ LH/FSH	↓ LH/FSH (↓ Estrogen/Testosterone)	
Uses	• Infertility (Anovulation/Oligospermia) • Delayed puberty	↓ Estrogen	↓ Testosterone
		• ER +ve breast Ca • Endometrial fibrosis • Endometriosis	• Prostate cancer : Goserelin (DOC)  • Precocious puberty

Drugs for Contraception

00:17:57

Regular Contraception :

Drug	MOA
Combined OCPs	 Ovulation
mini pills/POP	• ↑ Cervical mucous viscosity • ↓ Sperm penetration •  Blastocyst implantation
mifepristone	Blastocyst detachment

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Emergency Contraceptive :

Drug	Dose
Levonorgestrel : DOC 	1.5 mg 1 tab within 3 days
Ulipristal : SPRM (Selective progesterone receptor modulator)	30 mg 1 tab within 5 days

Growth Hormone Related Drugs

00:18:56

Classification & drugs	uses	Side effects
GH analogs : Somatrem/Somatropin	Dwarfism : - Turner's syndrome - Prader Willi syndrome - Noonan syndrome	mnemonic : CHILDREN - Carpal tunnel syndrome - Hyperglycemia - ICT is raised - Leukemia - Diabetes - c/i : - Retinopathy in DM - Neoplasia
GH releasing hormone (GHRH analog) : - Sermorelin - macimorelin - Tesamorelin	- Diagnosis of dwarfism. - AIDS related lipodystrophy (Tesamorelin)	-
Somatostatin analogs : - Octreotide LAR - Lanreotide - Pasireotide } Long acting	- Route : s/c - DOC for : Acromegaly  - Secretory diarrhea. - Glucagonoma/VIPoma/Somatostatinoma. - Other uses : - Thyrotrope adenoma. - Acute variceal bleeding.	- Hypothyroidism Gall stones
GH receptor antagonist : Pegvisomant	Drug resistant acromegaly	↑ Size of pituitary adenoma (monitor : MRI)

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Note :
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Gall stones also caused by fibrates.

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Steroid Hormone Related Drugs

00:28:34

	Potency	Half-life	uses
Hydrocortisone (Similar to cortisol)	Glucocorticoids (GC) $\uparrow \times 1$ mineralocorticoid (MC) $\uparrow \times 1$	8 - 12 hours	DOC for replacement  : • Addison's disease • Congenital adrenal hyperplasia (CAH)
Prednisone/ Prednisolone	GC $\uparrow \times 4$ MC $\uparrow \times 0.8$	12 - 36 hours	- • $\downarrow$ Inflammation • $\downarrow$ Immunity
methylprednisolone	GC $\uparrow \times 5$ MC $\uparrow \times 0.8$		
Triamcinolone	GC $\uparrow \times 5$ MC = 0		
Dexamethasone/ Betamethasone	GC $\uparrow \times 30$ MC = 0	36 - 72 hours	• Surfactant maturation • CAH pregnancy

Note :

Triamcinolone, Dexamethasone & Betamethasone :

- Pure glucocorticoids.
- No effect on BP.

Thyroid Related Drugs

00:33:09

Hyperthyroidism :

	MOA	uses	Side effects
Propylthiouracil 	Inhibits thyroid peroxidase $\downarrow$  Formation of T_3, T_4	DOC  for : • Hyperthyroidism in 1 st trimester. • Thyroid storm	Hepatotoxic
Carbimazole  (Prodrug), methimazole		DOC  for hyperthyroidism : Overall & 2 nd & 3 rd trimester	Teratogenic : • Choanal/esophageal atresia • Cutis aplasia  Cutis aplasia (Scalp defect)
Potassium iodide, Lugol's iodine	• $\ominus$ Thiol endopeptidase $\downarrow$ $\ominus T_3, T_4$ release. • $\downarrow$ Thyroid size. • $\downarrow$ vasculogenesis.	Pre-operative preparation of thyroid gland for surgery : To make thyroid gland firm & $\downarrow$ bleeding	-

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	MOA	uses	Side effects
Propranolol, Propylthiouracil, Amiodarone, Steroids	Inhibit peripheral conversion of T_4 to T_3	-	-
Radioactive iodide (I^{131}) (I^{123} : used in imaging)	Destroys follicular cells (Thyroid ablation) by β & γ rays	- Thyroid cancer - Recurrent Grave's disease - Hyperthyroidism in elderly	- Secondary cancer - Permanent hypothyroidism - C/i : Pregnancy

Note :

First drug given in thyroid storm (To prevent AF) : Propranolol.

Hypothyroidism :

	MOA	uses	Side effects
Levothyroxine	T_4 salt (Long acting)	- Oral route (On empty stomach) : - DOC  : Replacement - Thyroid cancer (To ↓ TSH) - IV route : myxedema coma	- Osteoporosis - Atrial fibrillation (↓ Dose used if pre-existing arrhythmia ⊕) - Thyrotoxicosis symptoms
Liothyronine	T_3 salt	- Oral route : T_3 given before radioactive iodine treatment in thyroid cancer - IV route : myxedema coma (T_4 + T_3 given)	-

AUTACOIDs

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Antihistaminics

00:00:10

H₁ Blockers :

	First generation	Second generation
Cross blood-brain barrier (BBB)	• Crosses (Sedating) • C/i in : - Elderly - Children - Drivers/pilots	Does not cross BBB (Non-sedating)
Drugs & their uses	• Muscarinic ⊖ : - Promethazine, Diphenhydramine, Dimenhydrinate - used in : a. EPS (Acute dystonia, Parkinsonism) b. motion sickness • Dicyclomine 🚫 : • Chlorpheniramine : - Least sedative 1st generation drug - use : Rx of cough (Daytime use) • Doxepin : - Currently under TCA • Cyproheptadine : - Currently under 5-HT_a ⊖ • Hydroxyzine	• Cetirizine : - metabolite of hydroxyzine - most sedative 2nd generation drug • Levocetirizine : more potent • Astemizole } Cause QT prolongation (Banned) • Terfenadine } • Fexofenadine : Overall least sedative (Daytime use) • Loratadine → Desloratadine (more potent) • Rupatadine : Additional anti-inflammatory effect • Topical antihistaminics : - Olapatadine - Levocabastine - Ketotifen - Azelastine (Preferred) : Intranasal spray for Rx of allergic rhinitis

Motion Sickness :

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1. Scopolamine :

- DOC.
- Transdermal patch applied the night before travel.

2. Promethazine/diphenhydramine/dimenhydrinate :

- Oral tablet.
- 1 hour before travel.
- Highly sedative.

Note :

DOC for allergic rhinitis : Intranasal steroids  (Fluticasone).

NSAIDs

00:05:41

Acetaminophen (Paracetamol) :

use : Analgesic & antipyretic effect.

Side effect : Hepatotoxic (m/c cause of poisoning worldwide).

- ↑ Risk in chronic alcoholics/fasting.
- HPE : Centrilobular necrosis + periportal sparing.
- Cause : metabolite NAPQI $\xrightarrow{\text{Depletes}}$ Glutathione.
- Toxic dose :
 - > 150 - 250 mg/kg or > 10 g : Hepatotoxic.
 - > 20 g : Fatal.
- Treatment of toxicity :
 - DOC : N-acetyl cysteine $\begin{cases} \rightarrow \text{Blocks NAPQI.} \\ \rightarrow \text{Replenishes glutathione.} \end{cases}$
 - If no response : Liver failure $\rightarrow$ Emergency liver transplantation.

Aspirin :

Dose dependent effects :

- 50 - 325 mg OD : Anti-aggregant.
- 325 - 650 mg SOS : Antipyretic/analgesic.
- 3 - 4 g/d (Divided dose) :
 - Anti-inflammatory.
 - E.g. : RA (Aspirin $\rightarrow$ Analgesic of choice).

Side effects :

- Reye's syndrome (Hepatic encephalopathy) : C/i in viral fever in children.
- ↑ uric acid : C/i in gout.

----- Active space -----

management of toxicity :

- Bicarbonates (Antidote of choice).
- If pulmonary edema developed : Dialysis.

Indomethacin :

uses :

- DOC for acute gout .
- Closure of patent ductus arteriosus (PDA).

Ibuprofen  :

DOC for closure of PDA in India.

Piroxicam :

Long acting $\xrightarrow{\text{used for}}$ Chronic pain.

Nimesulide :

Hepatotoxic : C/i in children < 12 years.

Ketorolac :

uses :

- Post-operative pain.
- Eye pain for ocular pain.

Selective COX-2 Inhibitors :

↑Risk of myocardial infarction : Not preferred.

Celecoxib, etoricoxib : 3rd line for pain and inflammation.

Parecoxib : Post operative pain.

Lumiracoxib : Hepatotoxic (Banned).

valdecoxib, rofecoxib : ↑Risk of MI (Banned).

Prostaglandin Analogs

00:13:24

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Prostaglandin	Analog	Uses
Prostaglandin E_1	misoprostol	- Abortion (+ mifepristone) - NSAID induced gastric ulcer - maintain patency of DA
	Alprostadil	- maintain patency of DA - Erectile dysfunction
Prostaglandin E_a	Dinoprostone	- Cervical ripening (DOC) - PPH
Prostaglandin $F_a\alpha$	Carboprost	PPH
	Latanoprost/bimatoprost	Open angle glaucoma
Prostaglandin I_a	- Epoprostenol (Synthetic) - Iloprost - Beraprost - Treprostinil } Analogs • Selexipag : Receptor agonist	Pulmonary HTN

Pulmonary HTN - Rx :

Stage II/III :

- Endothelin $\ominus$: Bosentan, ambrisentan (DOC).
- Alternative : PDE-5 $\ominus$ (Sildenafil); CCB.

Stage IV : Epoprostenol (DOC).

PPH

DOC : Oxytocin .

Disease Modifying Anti-Rheumatoid Drugs (DMARDs)

00:15:48

Conventional DMARDs	Biological DMARDs	Targeted DMARDs
1. methotrexate (DOC) : - MOA : $\uparrow$ Adenosine - Long term s/e : Cirrhosis (Seen in rheumatic/psoriatic arthritis) 2. Hydroxychloroquine 3. Sulfasalazine 4. Cyclophosphamide 5. Immunomodulators : - Azathioprine - Cyclosporine - mycophenolate mofetil - Leflunomide	Route : Parenteral 1. IL-1 $\ominus$: Anakinra 2. CD-20 $\ominus$: Rituximab 3. TNF- α $\ominus$: mnemonic : In CAGE - Infliximab - Certolizumab - Adalimumab - Golimumab - Etanercept 4. IL-6 $\ominus$: - Sarilumab - Tocilizumab 5. CD-80/86 $\ominus$: Abatacept	- Last line drugs : JAK inhibitors - Route : Oral 1. Baricitinib 2. Upadacitinib 3. Tofacitinib

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Anti-gout Drugs

00:18:53

Acute Gout :

-  DOC : Indomethacin.
- Alternative : Colchicine →  microtubules →  chemotaxis.

Chronic Gout :

Aim : ↓ uric acid (UA).

Xanthine oxidase inhibitors :

- MOA :  UA synthesis.
- Drugs :
 1. Allopurinol (DOC) .
 2. Oxypurinol (If SJS with allopurinol).
 3. Febuxostat (most effective).
- Side effects :
 - Xanthine stones.
 - Steven Johnson syndrome (HLA-B5801).
 - Acute gout (Prophylaxis with indomethacin)

Uricosuric drugs :

- MOA : ↑ UA excretion.
- Drugs :

1. Sulfapyrazone	} monotherapy or add-on to allopurinol.	} effective in renal failure.
2. Probenecid		
3. Benzbromarone		
4. Lesinurad : Only as add-on		
- S/e : urate stones.

Uricase analogs :

- MOA : metabolize UA.
- $$\text{UA} \xrightarrow{\text{uricase (Oxidase)}} \text{5-hydroxy isourate} \xrightarrow{\text{Hydrolysis}} \text{Allantoin (Water soluble).}$$
- Drugs :
 - Pegloticase (Last line).
 - Rasburicase : Used in tumor lysis syndrome.

DRUGS ACTING ON RS, GIT AND BLOOD

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Antiasthmatic Drugs

00:00:14

Drugs	MOA/use	Side effects
- Beta-2 agonists - Anticholinergics	(Refer to 'Drugs acting on autonomic nervous system')	
methylxanthines : - Theophylline - Aminophylline	- Bronchodilation : - Adenosine A₁ ⊖ - PDE-3 > 4 ⊖ - Anti-inflammatory : - PDE-4 ⊖ - Histone deacetylase ⊕ (Similar to steroids) ↑ IL-10	- Adenosine A₁ ⊖ : Seizure, arrhythmia, diuresis - PDE-4 ⊖ : GI symptoms, headache
Inhaled corticosteroids (ICS) : - Fluticasone (most potent) - mometasone - Ciclesonide - Beclomethasone } Soft steroids Note : Soft steroids : - Prodrugs; activated in the lungs ↓ Risk of oral candidiasis	- DOC for Persistent BA - TOC for intermittent BA : ICS + Formoterol as reliever	- Hoarseness of voice (m/c) - Oropharyngeal candidiasis (most severe s/e)
Systemic steroids	Acute exacerbation of BA : ↑ β ₂ receptor density in bronchi → ↑ Agonist effect	-
LOX ⊖ : Zileuton	Add on drugs	Hepatotoxic : Zileuton > montelukast
LT C ₄ /D ₄ ⊖ : montelukast		
mast cell stabilizers : - Cromolyn sodium - Nedocromil	Add on drug	Safest drugs
Anti-IgE : Omalizumab	Severe persistent asthma	Not effective in atopic dermatitis (D/t ↑ Ig-E)
Anti IL-5 : Reslizumab, mepolizumab	Severe eosinophilic asthma	-
Anti IL-5 R : Benralizumab		
Anti IL-4 R : Dupilumab		

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Mx of Acute Asthma Attack :

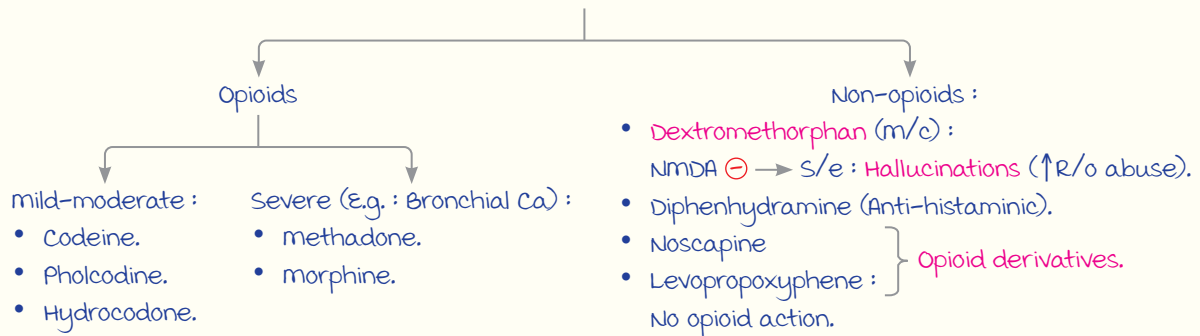
- ICS + Formoterol (1 pump).
- Salbutamol $\xrightarrow{5 \text{ min}}$ ICS (When taken separately).
- Salbutamol $\xrightarrow{1 \text{ min}}$ Repeat (If symptoms not relieved).

Antitussives

00:05:52

1. Dry Cough :

Centrally acting agents :

2. Productive Cough :

Expectorant : Guaifenesin.

MOA : Irritation of gastric mucosa $\rightarrow$ Vagal $\oplus$ $\rightarrow$ Bronchial $\oplus$ $\rightarrow$ Expectoration of mucus.

mucolytics : Liquify mucous.

- N-acetyl cysteine : Break disulfide bond.
- Ambroxol/bromhexine : Depolymerize mucopolysaccharide.

mx : Syrup guaifenesin + Ambroxol/Bromhexine + Salbutamol + menthol.

3. New Drugs in COPD :

- Roflumilast : PDE-4 $\ominus$.
- Ensifentrine : PDE-3 & PDE-4 $\ominus$.

Drugs Used in Peptic Ulcer Disease (PUD)

00:10:36

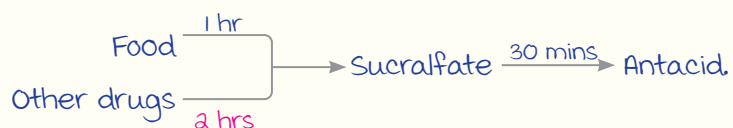
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Aim : Ulcer healing.

Drugs	MOA	uses	Side effects
misoprostol	Gastroprotective	NSAID induced gastric ulcer (most specific)	Abdominal cramps
Sucralfate	In acidic pH : Sucralfate ↓ Polymerizes Sticky & viscous ↓ Coats ulcer	- PUD - Rectal ulcer	- Constipation - Gastric bezoars ↓ Absorption of other drugs
Bismuth subsalicylate/ subcitrate	Gastroprotective	H. pylori - Traveller's diarrhoea	- Constipation - Black discoloration of stool & tongue
Antacids : Salts of Al & mg	- Neutralise acid - Al → Constipation - mg → Diarrhea	- Dyspepsia (m/c) - PUD - GERD	↓ Absorption of other drugs
Proton pump inhibitors (PPI) : - Omeprazole - Esomeprazole (Longer acting) - Pantoprazole - Lansoprazole - Rabeprazole	⊖ Food induced acid secretion (Taken 30 min before food) - Irreversible inhibition	- PUD - GERD - Zollinger Ellison Syndrome H. pylori - Barrett's esophagus (Lifelong Rx)	- GI upset - Pneumonia - Pseudomembranous enterocolitis - Deficiency : Iron, calcium, vit. B₁₂ - Osteoporosis (Hip fracture) - Hypergastrinemia
vonoprazan	- Potassium competitive acid blocker (PCAB) ↓ Basal + Food induced acid secretion (Taken with/without food)	H. pylori	-
H ₂ blockers : - Cimetidine - Ranitidine - Famotidine	↓ Basal acid secretion	DOC for : Prophylaxis of aspiration pneumonia in post-op patients	Cimetidine : - Impotence - Gynaecomastia - Galactorrhoea

Note :

Spacing of sucralfate :



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Prokinetics

00:18:03

Drugs		use	Side effects
D ₂ antagonists	metoclopramide	• GERD • Anti-emetics	• EPS : Acute dystonia • Hyperprolactinemia
	Domperidone		No EPS
5-HT ₄ agonists : • mosapride • Itopride • Prucalopride		• GERD • Constipation	↑QT : Cisapride, Tegaserod (Banned)
motilin receptor agonists : • erythromycin • mitemincal		Gastroparesis	-
CCK I/A receptor antagonists : Dexloxiplumide			

Antiemetics

00:19:25

Drugs for Chemotherapy Induced Nausea/Vomiting :

Group	Drugs
5-HT ₃ antagonists (DOC) 	• Ondansetron (Shortest) • Palonosetron (Longest, most potent)
NK1R antagonists	• Aprepitant • Rolapitant • Netupitant } Longer acting
CB1R agonists	• Dronabinol • Nabilone } s/e : Hypotension, blood shot eyes
Others	• Dexamethasone • metoclopramide • Olanzapine

5-HT₃ Antagonists :

Other uses :

- DOC for : 
 - Post op vomiting.
 - Radiation induced vomiting.
- morning sickness.

s/e : Hypotension.

Note :

DOC for morning sickness : Doxylamine .

Laxatives

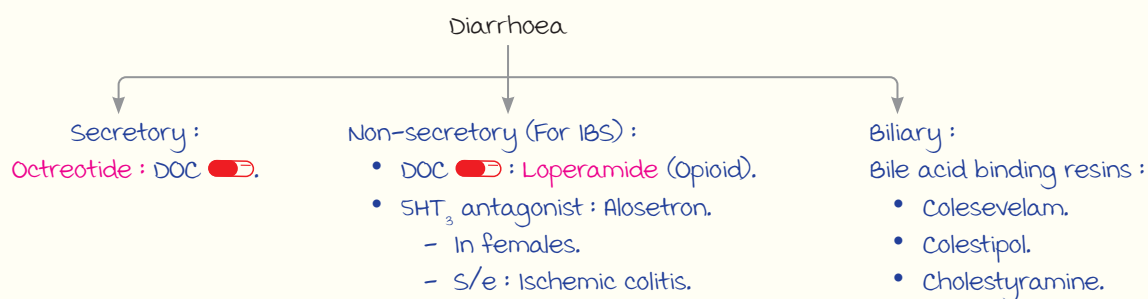
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↑Water in intestine	↑Intestinal contraction	↑Stool bulk	↑Stool softening (Least effective)
1. Osmotic laxatives : - mannitol : • 2 nd line for constipation (D/t flatulence & ↑ cost) • Hepatic encephalopathy ($\text{NH}_3 \rightarrow$ Polar ammonium ion) - Polyethylene glycol (PEG) DOC : IBS + constipation - NaPO_4 : used for bowel preparation	1. Bisacodyl, senna, cascara : - Low grade inflammation of large intestine $\rightarrow$ ↑ Contraction. - Effect : After 6 - 8 hrs (Night dose) - max 10 days (Short term Rx) : S/e : Atonic colon - Senna : a. melanosis coli b. Discolouration of urine	1. Probiotics (Beneficial microbes) : - Lactobacillus - Saccharomyces - B. clausii	1. Docusate sodium
2. Chloride secretory agents : - Lubiprostone : (⊕ Type II chloride channels) - Linaclotide : (⊕ Guanylate cyclase) ↑ cGMP $\rightarrow$ ⊕ CFTR	2. 5-HT ₄ agonists : - mosapride - Prucalopride	2. Prebiotics (Dietary fibers) : - methylcellulose - Psyllium husk - Bran	2. Docusate calcium
3. Tenapanor : ⊖ Na^+ -proton exchanger			

Anti-Diarrhoeal Agents

00:25:01



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Antiaggregants

00:25:54

MOA	Drugs		Uses	Side effects
COX-I inhibitor → ↓ Thromboxane A ₂ (TXA ₂) synthesis	Aspirin		Primary prophylaxis : MI, stroke	Bleeding
Protease activated receptor-1 (PAR-1) blocker	Vorapaxar		Primary prophylaxis of MI	Intracranial bleed (C/I in cerebrovascular d/s : stroke, TIA)
GP IIb-IIIa blockers	• Abciximab • Tirofiban • Eptifibatide		• PCI in MI • Unstable angina	• Bleeding • Thrombocytopenia
Adenosine diphosphate (ADP)/ P _a γ _{1a} blocker	Irreversible (m/c used)	Clopidogrel (m/c used)	Primary prophylaxis : MI, stroke (2 nd line d/t CYP2C19 polymorphism in some population)	-
		Ticlopidine	-	Toxic
		Prasugrel (most potent)	PCI in MI	Intracranial bleed (C/I in cerebrovascular d/s : Stroke, TIA)
	Reversible	Cangrelor (Adenosine analog, IV, short acting)	PCI in MI	-
		Ticagrelor (Oral, long acting)	• Prophylaxis of MI & stroke • Acute coronary syndrome	-

Anticoagulants

00:28:33

Direct Thrombin Inhibitors (DTI) :

MOA		Drugs	Antidotes	Coagulation monitor	Uses
 Direct acting oral anticoagulants (DOAC)	Oral direct thrombin inhibitors (DTI)	Dabigatran	Idarucizumab	Not required	• DVT Rx • DOC for prophylaxis of : - DVT (long term) - Thrombosis in non-valvular AF (Native valve)
	Oral Xa inhibitors	• Apixaban • Edoxaban • Rivaroxaban	Andexanet alfa		
• Blocks VKOR • ↓ Factors II, VII (1st to ↓), IX, X, Protein C (2nd to ↓) & S		Warfarin 	• 4 factor prothrombin complex > FFP • Vitamin K (Improves outcome)	PT/INR	• Rx & prophylaxis of DVT • DOC for prophylaxis of thrombosis in : - Valvular AF (mechanical valve) - Non-valvular AF with severe mitral stenosis (< 1.5 cm²)

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MOA	Drugs	Antidotes	Coagulation monitor	Uses
Parenteral direct thrombin inhibitors (PE DTI)	- Argatroban - Desirudin - Bivalirudin - Lepirudin	-	aPTT	Heparin induced thrombocytopenia (HIT)

S/e of warfarin :

- Skin necrosis : D/t rapid ↓ in protein C > S.
- Purple toe.
- Teratogenic :
 - mid facial hypoplasia (Flat nose).
 - Stippled epiphyseal calcification.
 - CNS defects.



Purple toe

Note :

- Drugs that do not require monitoring :
 - DOAC.
 - LMWH.
 - Fondaparinux.
- Drugs requiring monitoring :
 - Warfarin : PT/INR.
 - PE DTI } aPTT.
 - UFH }

Indirect Thrombin Inhibitors (ITI) :

	unfractionated heparin (UFH)	LMWH (Enoxaparin, tinzaparin)	Fondaparinux
Size	Large	Small	Smallest
Action	Factor Xa = IIa	Factor Xa > IIa	Factor Xa only
Risk of HIT	High	Low	Absent
Route	- S/c : Prophylaxis - IV : Rx	S/c : Prophylaxis & Rx	
T _{1/2}	Short (Physiologically synthesized by mast cells & consumed by macrophages)	Long acting	
Dosing	multiple	OD	

----- Active space -----

	unfractionated heparin (UFH)	LMWH (Enoxaparin, tinzaparin)	Fondaparinux
Excretion	Absent (Cleared by macrophages)	unchanged in urine	
In renal failure	used	c/i	
Antidote	Protamine sulphate	-	
monitoring	aPTT	Not required	
uses	-	DOC for post-op prophylaxis of DVT	Rx of HIT d/t UFH/LMWH (Argatroban : DOC)
		DOC for Rx of thrombosis : DVT, pulmonary embolism	

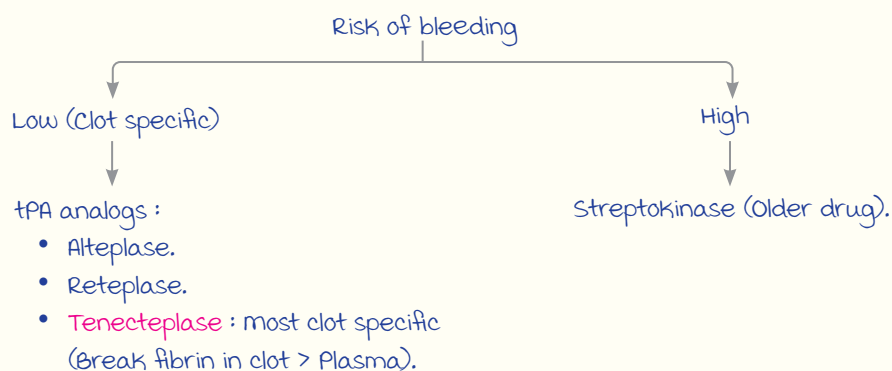
Fibrinolytics/Thrombolytics

00:39:12

Physiology :

Plasminogen $\xrightarrow{\text{Tissue plasminogen activator (tPA)}}$ Plasmin $\longrightarrow$ Breaks fibrin (In plasma & clots).

Drugs :



c/i : NSTEMI (D/t risk of bleeding > benefit).

use : STEMI.

s/e : Bleeding (DOC : Tranexamic acid > epsilon-aminocaproic acid (EACA))

Hematopoietic Agents

00:41:45

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	Drugs		uses	s/e
Erythropoiesis	Erythropoietin analogs : • Epoetin alpha • Darbopoetin 		Anemia d/t : • CRF • Chemotherapy • Zidovudine • Dialysis	• Pure red cell aplasia • Hypertension • Thrombosis • Iron deficiency
Granulopoiesis	G-CSF analogs : • Lenograstim • Filgrastim • Lipegfilgrastim  Given once → (Long acting)		• Neutropenia • HIV • Chemotherapy cycle	Bone pain
	GM-CSF Analogs : Sargamostim			
Thrombopoiesis	IL-11 analog : Oprelevakin		Chemotherapy induced thrombocytopenia	-
	Thrombopoietin agonist	• Eltrombopag • Romiplostim	Immune thrombocytopenic purpura (ITP)	
		Newer drugs  : • Lusutrombopag • Avatrombopag	↓ R/o bleeding in patients of liver cirrhosis prior to planned procedures	

G-CSF : Granulocyte-colony stimulating factor.

GM-CSF : Granulocyte monocyte-colony stimulating factor.

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IMMUNOMODULATORS AND ANTICANCER DRUGS

Immunosuppressants

00:00:05

	MOA	uses	s/e	
- Cyclosporine  - Tacrolimus	Calcineurin inhibitors : ↓ Transcription of IL-2	1 st line drugs post transplantation (GVHD)	Tacrolimus > Cyclosporine : - Nephrotoxicity - Neurotoxicity - Hyperkalemia - Hypertension - Hepatotoxicity - Hyperglycemia	Cyclosporine : - Hirsutism - Hyperplasia of gums - Hyperlipidemia - Hyperuricemia
Everolimus Sirolimus	m-tor inhibitors	2 nd line drugs	Hypokalemia	
Azathioprine	Prodrug of 6-mercaptopurine	-	- Bone marrow suppression - Hepatotoxicity	
Belatacept	CD-80/86 inhibitor	-	-	
Alemtuzumab	CD-52 inhibitor	-	-	
Belumosudil	Rho kinase inhibitor	-	-	
Thalidomide (Derivative : Lenalidomide)	- Anti-inflammatory - Anti-angiogenic - Anti-neoplastic - Immunosuppressant	- Rheumatoid arthritis - multiple myeloma - GVHD	Phocomelia : D/t cereblon & tubulins block <pre> graph TD A[Phocomelia : D/t cereblon & tubulins block] --> B[Block vasculogenesis] A --> C[↑ Free radicals] B --> D[Block fetal organ development] C --> D </pre>	
Basiliximab	Blocks IL-2 receptor/ CD-25	Prophylaxis of acute graft rejection	-	
mycophenolate mofetil 	Blocks inosine monophosphate (IMP) dehydrogenase	1 st line immunosuppressant	GIT upset	
Leflunomide	Blocks dihydroorotate dehydrogenase (Inhibits pyrimidine synthesis)	-	- Bone marrow suppression - Hepatotoxic	
muromonab	Blocks CD-3	- Rx of acute graft rejection - Rebound hypertension	-	

Note :

Azathioprine/6-mercaptopurine + Allopurinol (Xanthine oxidase $\ominus$)
(metabolized by x.o.)

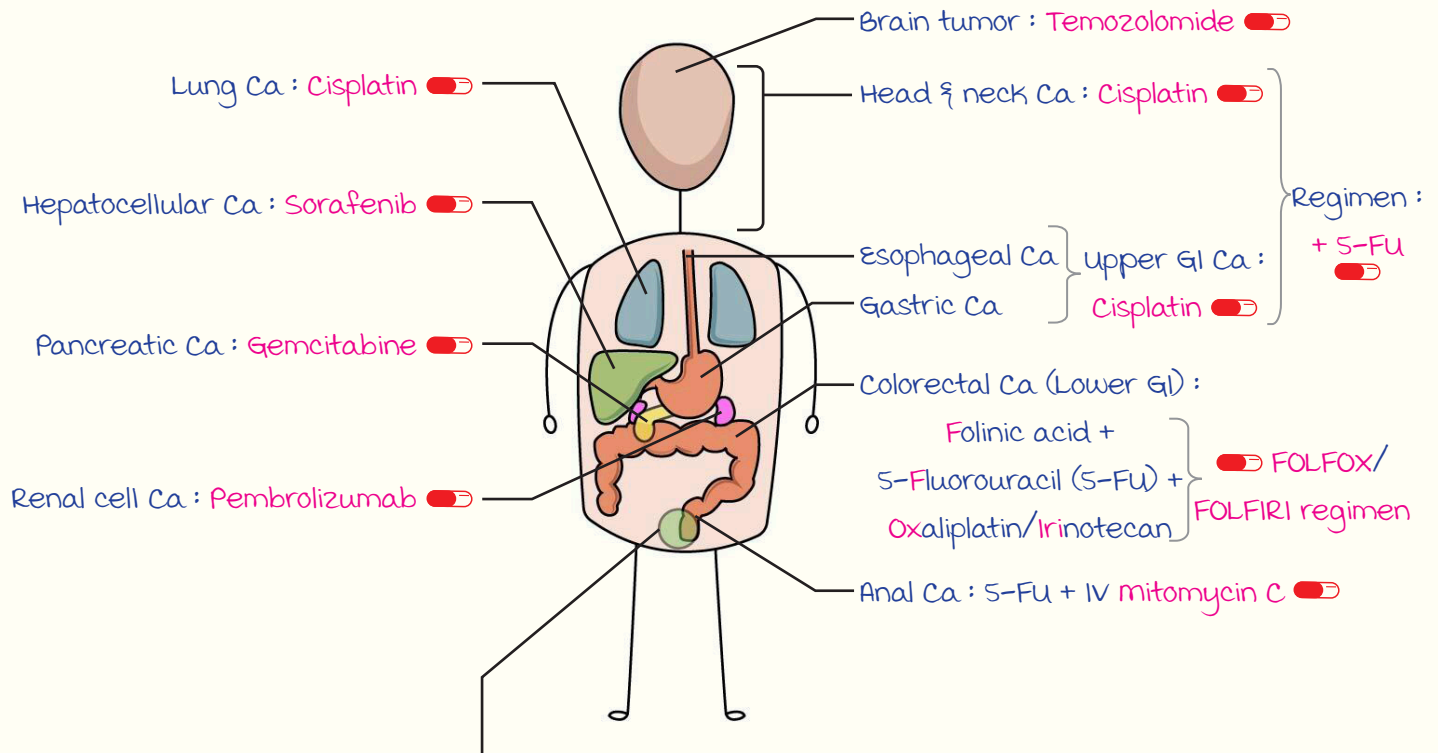
↓ Dose of azathioprine/6-mercaptopurine by 75%.

- Abatacept : Rx of rheumatoid arthritis.

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Solid Tumors

00:06:43



urogenital Ca (DOC : Cisplatin)	
Cancer	Add on Rx (for regimen)
uterine/ ovarian Ca	Paclitaxel
Bladder Ca	Gemcitabine
Cervix Ca	- (Only Cisplatin)
Testicular Ca	BEP (Bleomycin + etoposide + cisplatin)

Prostate Ca :
Goserelin (Exception)

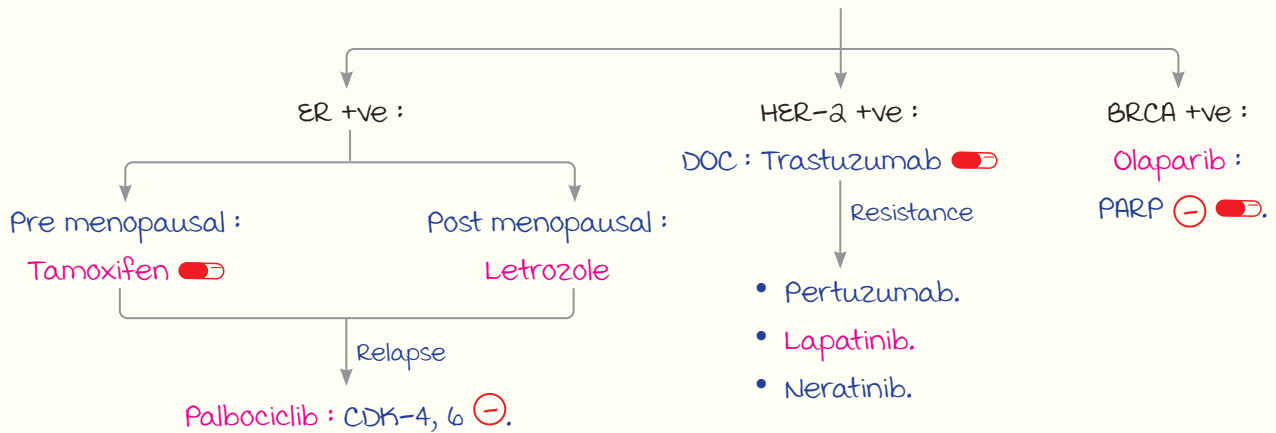
- Choriocarcinoma
 - Ectopic pregnancy
 - Osteosarcoma
- DOC : methotrexate

----- Active space ----- mitomycin C  :

- IV : Anal Ca.
- Intravesical : Bladder Ca.
- Topical : Prevent laryngotracheal stenosis/post nasal surgery synechiae.

Breast Cancer

00:14:20



Leukemia

00:17:32

	Regimen
ALL	VPAD regimen : • Vincristine • Prednisolone • Asparaginase • Daunorubicin
AML	Cytarabine + Daunorubicin/Idarubicin ↓ if FLT ₃ mutation + Gilteritinib/midostaurin
CLL	• Fludarabine • Cyclophosphamide • Rituximab
CML	BCR-ABL Tyrosine Kinase  • 1st gen : Imatinib : DOC  ↓ NR • 2nd gen : - Dasatinib - Nilotinib - Bosutinib ↓ Resistance • 3rd gen : Ponatinib ↓ Resistance • Asciminib : BCR-ABL TK  (Allosteric site ) • Omacetaxine : BCR-ABL protein  • IFN-α } Enzymatic site
Hairy cell leukemia	Cladribine : DOC 
Promyelocytic leukemia	Retinoic acid : DOC  → Arsenic trioxide

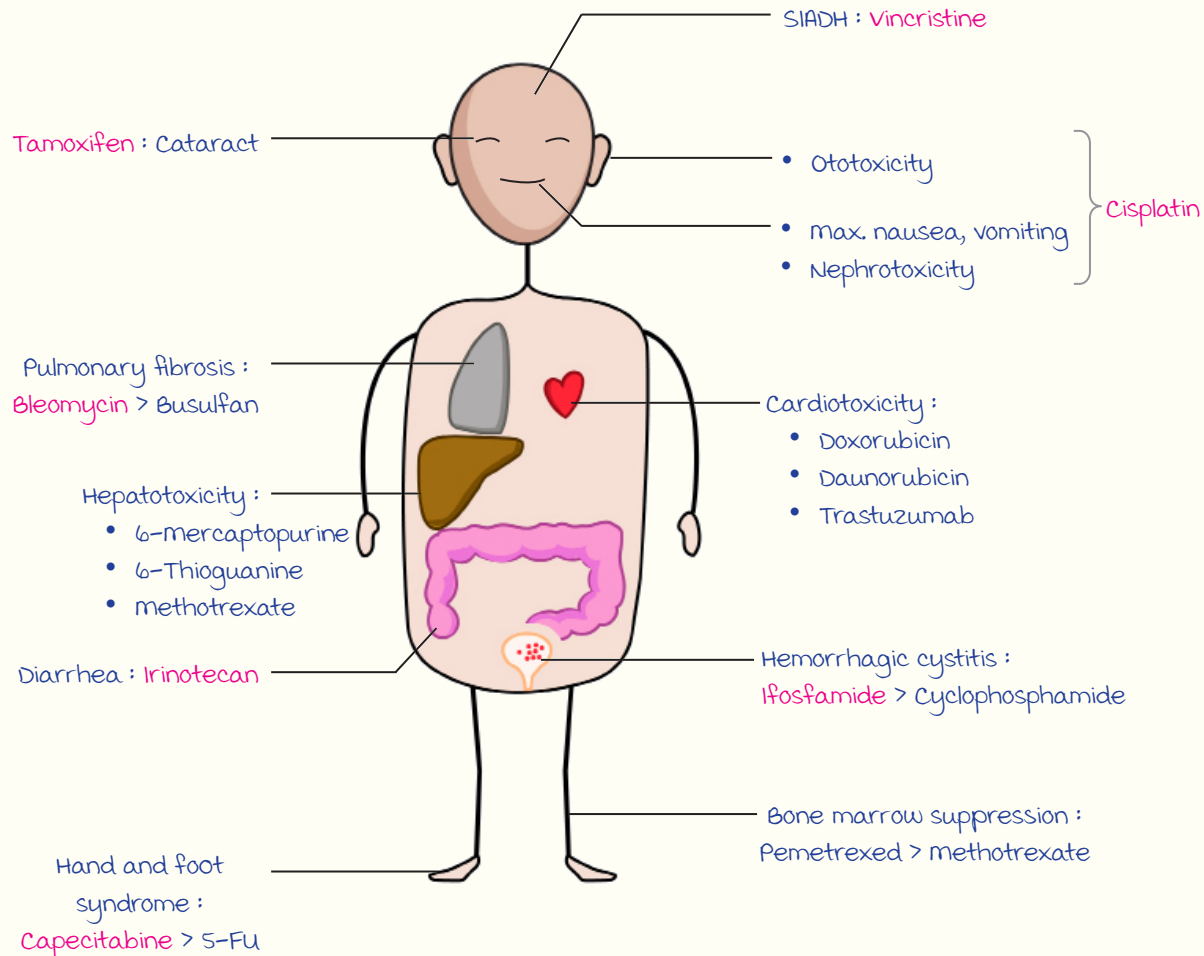
Note :

- Sickle cell disease : **Hydroxyurea** DOC
- ↓
- MOA : $\uparrow$ HbF.
- S/e : Painful leg ulcers.
- GIST : Imatinib DOC

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Side Effects and Antidotes of Anti-Cancer Drugs

00:22:15



Hand and foot syndrome

Bleomycin :
Flagellate dermatitis

----- Active space -----

Antidotes :

Toxicity	Drug	Antidote	
Hand and foot syndrome	5-FU	Vit B ₆	
mucositis & bone marrow suppression	• methotrexate • Pemetrexed	Folinic acid (Leucovorin)	-
		> Folic acid	+ Inj. Vit B ₁₂
Cardiotoxicity	Anthracyclines (Doxo/Daunorubicin)	Dexrazoxane	
Hemorrhagic cystitis	• Ifosfamide • Cyclophosphamide	mesna	
Hyperuricemia d/t tumor lysis syndrome	-	• Allopurinol (Solid tumors) • Rasburicase (Leukemias)	
Neutropenia	-	Lipegfilgrastim	
Thrombocytopenia	-	Oprelvekin	
Anemia	-	Darbepoetin	
Delayed diarrhea	Irinotecan	Loperamide	
Nephrotoxicity	Cisplatin	NaCl + mannitol	

Miscellaneous Anti-Cancer Drugs

00:28:36

Nomenclature of monoclonal antibodies :

- Trastuzumab - tu/tum : Tumor.
- Abciximab - c/ci : Circulation.
- Bezlotoxumab - tox : Toxin (C. difficile).
- Palivizumab : Virus.
- Infliximab - li/lim : Lymphocyte (RA).
- Erenumab - n/ne : Neurology.
- Raxibacumab - bac : Bacteria (Anthrax).
- Denosumab - os : Osteoporosis.

Kinase inhibitors :

1. Imatinib : BCR-ABL TK ⊖.
 2. Gilteritinib : FLT3 Kinase ⊖ (AML).
 3. Dabrafenib : BRAF ⊖
 4. Cobimetinib : MEK 1/2 ⊖
- } melanoma.
5. Idelalisib : PI-3 K ⊖ (Leukemia).
 6. Ibrutinib : Bruton's TK ⊖ (B-cell lymphomas).

Asparaginase : Leukemia.

Olaparib :

- Poly-ADP ribose polymerase (PARP) ⊖.
- For Ca breast.

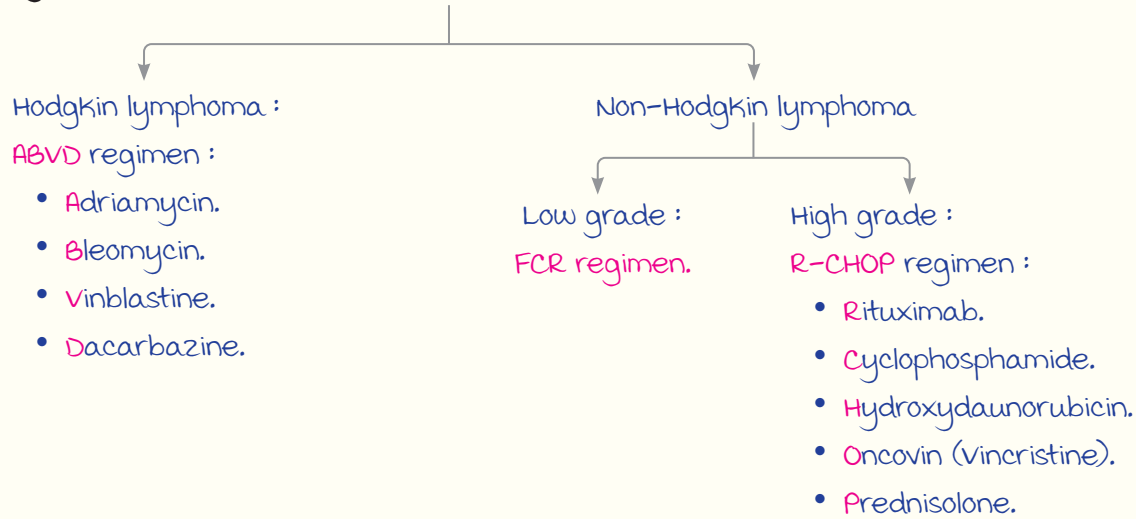
Palbociclib : CD-4/6 ⊖.

Bortezomib :

- Proteasome ⊖.
- DOC ⊕ for multiple myeloma along with Lenalidomide + dexamethasone.

Lymphoma :

----- Active space -----

Immune-checkpoint Inhibitors :